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Perceptual impact of electric vehicle acoustic alerting systems on urban soundscapes: using the ISO 12913 framework

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Abstract:	With the increasing number of electric vehicles (EVs), their Acoustic Vehicle Alerting System (AVAS) is becoming an essential element shaping future urban acoustic environments. This study, therefore, investigates how AVAS sounds may influence urban soundscapes, using the ISO 12913 framework. A total of 43 AVAS sounds from 17 EVs were recorded and evaluated under laboratory conditions, focusing on two main perceptual attributes: pleasantness and eventfulness. In parallel, psychoacoustic parameters were analyzed to investigate associations with soundscape attributes. Although linear relationships were mostly insignificant, nonlinear trends emerged between soundscape attributes and psychoacoustic parameters. Cluster analysis based on soundscape scores identified a group of EVs, characterized by AVAS features dissimilar to conventional engine sounds, with notably higher pleasantness ratings. Overall, the study provides new evidence that AVAS sounds can influence urban acoustic environments, highlighting their potential as a design tool for enhancing future urban soundscape quality.
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Cover Letter

March 29, 2026

Prof. James F. Lynch
Editor-in-Chief

Prof. Francesco Aletta
Associate Editor

The Journal of the Acoustical Society of America

Dear Professors James F. Lynch and Francesco Aletta:

We would like to submit the manuscript entitled “**Perceptual impact of electric vehicle acoustic alerting systems on urban soundscapes: using the ISO 12913 framework**” for publication in *The Journal of the Acoustical Society of America*. The transition to the era of electric vehicles (EVs) is bringing substantial changes to the urban sound environment. Since EVs produce no engine noise, they can contribute to reducing urban traffic noise compared to conventional internal combustion engine (ICE) vehicles. However, EVs are required by law to use an Acoustic Vehicle Alerting System (AVAS) to warn pedestrians of their approach during low-speed driving. As a result, AVAS sounds have emerged as a new element of traffic noise and are expected to reshape the soundscape of urban environments.

In light of this, this study aims to evaluate the perceptual characteristics of AVAS sounds based on a soundscape assessment framework and to examine how these sounds influence the urban soundscape. AVAS signals are designed to enhance pedestrian detectability through changes in sound pressure level and frequency shifting according to speed, and they vary significantly depending on vehicle brands. In this study, various AVAS sounds were classified into clusters using a soundscape Pleasantness-Eventfulness model and investigated the characteristics of each cluster.

To this end, we collected AVAS sounds from 17 different EV models. Considering

that AVAS sounds vary with vehicle operation, we recorded sounds in three drive conditions: stationary with the drive gear engaged, pass-by at 20 km/h, and reverse at 6 km/h. The results show that soundscape evaluation outcomes differ according to drive condition. Psychoacoustic parameters of AVAS sounds were analyzed, and five representative indicators were extracted by principal component analysis. Correlations between these indicators and soundscape evaluation scores were also explored, revealing nonlinear tendencies between psychoacoustic parameters and perceptual evaluations

Interestingly, the results also revealed that EVs emitting AVAS sounds dissimilar to typical ICE engine noises were evaluated as having more pleasant soundscapes compared to others. This suggests that as EVs replace ICE vehicles in urban traffic, the overall perception of traffic-related soundscapes may shift toward a different qualitative character.

We believe that this article will be of interest to readers of *The Journal of the Acoustical Society of America*, as it provides valuable insights into the evolving urban acoustic environment in the context of widespread EV adoption. We confirm that this manuscript has not been published and is not under consideration for publication elsewhere. The authors declare that there are no conflicts of interest. Thank you for your consideration of this manuscript.

Sincerely,

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Perceptual impact of electric vehicle acoustic alerting systems on urban soundscapes: using the ISO 12913 framework

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With the increasing number of electric vehicles (EVs), their Acoustic Vehicle Alerting System (AVAS) is becoming an essential element shaping future urban acoustic environments. This study, therefore, investigates how AVAS sounds may influence urban soundscapes, using the ISO 12913 framework. A total of 43 AVAS sounds from 17 EVs were recorded and evaluated under laboratory conditions, focusing on two main perceptual attributes: pleasantness and eventfulness. In parallel, psychoacoustic parameters were analyzed to investigate associations with soundscape attributes. Although linear relationships were mostly insignificant, nonlinear trends emerged between soundscape attributes and psychoacoustic parameters. Cluster analysis based on soundscape scores identified a group of EVs, characterized by AVAS features dissimilar to conventional engine sounds, with notably higher pleasantness ratings. Overall, the study provides new evidence that AVAS sounds can influence urban acoustic environments, highlighting their potential as a design tool for enhancing future urban soundscape quality.

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I. INTRODUCTION

The global adoption of electric vehicles (EVs) has accelerated rapidly (IEA, 2025), making them an increasingly common mode of transportation in everyday life. This transition is bringing wide-ranging impacts across transportation systems³⁶, environmental conditions²⁰, and broader social contexts⁵⁴. One area that is particularly affected is the urban sound environment^{8,30,31}.

Unlike internal combustion engine (ICE) vehicles, EVs operate without engine noise, potentially reducing traffic-related noise pollution^{9,18,39,43}. However, the absence of engine sound also introduces safety concerns, as pedestrians may struggle to detect the approach of a moving vehicle^{34,50}. To address this issue, many countries have mandated the installation of Acoustic Vehicle Alerting Systems (AVAS). These systems are designed to emit artificial sounds from external speakers at low speeds—typically under 30 km/h—so that pedestrians can perceive the presence of EVs^{9,16}.

The countries including South Korea, the United States, members of the European Union, and China have enacted regulations mandating the installation of AVAS in electric vehicles^{16,18,44}. In response, research into the design and development of AVAS sounds has recently gained significant momentum^{34,41,47}. Previous studies on AVAS^{1,17,40} have primarily focused on optimizing acoustic features, such as frequency characteristics and sound patterns across different drive conditions, to enhance pedestrian detectability. Meanwhile, Fagerlönner et al.¹⁴ explored AVAS design strategies in relation to user acceptance, including consumer satisfaction and perceived usefulness. They also highlighted the necessity for manufacturers to prioritize sound design from the public's perspective.

However, as the adoption of EVs continues to rise, it is also important to consider that AVAS is emerging not only as a safety device for pedestrians but as a new component of the urban acoustic environment³¹. AVAS sounds of EVs possess noise reduction potential compared to ICE vehicles³⁰. However, Laib and Alexander Schmidt³⁰ mentioned that the

overlay of AVAS with varying tone pitches and tone durations can lead to a cacophony, resulting in a disharmonic urban soundscape. Therefore, it has been suggested that AVAS sounds should be designed with explicit consideration of the urban acoustic environment^{14,30}. Despite growing concerns regarding the potential impact of AVAS on urban soundscapes, empirical research on this issue remains limited.

Soundscape refers to the acoustic environment as perceived or experienced by individuals or groups within a specific context, which emphasizes the perceptual construct of sound environments²³. As one of the dominant components of urban soundscapes, traffic noise is expected to undergo significant changes with the increasing prevalence of EVs.^{39,47}. While EVs contribute to a reduction in overall traffic noise levels, the introduction of AVAS sounds to ensure pedestrian safety has introduced new auditory elements to the urban environment. These artificial sounds alter not only the acoustic characteristics but also the perceptual and emotional qualities of the soundscape. Therefore, understanding the perceptual and affective impacts of AVAS sounds is crucial for achieving a balanced urban auditory experience that supports both safety and environmental well-being. In this regard, this study aims to examine the relationships between the physical attributes of AVAS sounds and listeners' subjective evaluations, as well as to characterize their perceptual and affective properties in urban contexts.

This study investigates how AVAS sounds in EVs may influence the urban sound environment by analyzing their psychoacoustic parameters and conducting perceptual evaluations. The study addresses four research questions: (1) Which psychoacoustic parameters serve as key components in characterizing EV AVAS sounds? (2) What are the perceived affective qualities of EV AVAS sounds? (3) How are psychoacoustic parameters associated with the perceived affective qualities of the soundscape? (4) How can EVs be categorized based on the soundscape attributes of their AVAS sounds? To answer these questions, AVAS sounds from 17 EV models were analyzed and evaluated using the ISO 12913 framework. The findings

provide insight into how AVAS sounds may contribute to the urban soundscape experience as EVs become more prevalent. This study also highlights the need for AVAS sound design that considers both pedestrian safety and environmental impact.

The research flow is presented in Fig. 1. Section 2 describes the experimental design in detail and explains the analytical approaches adopted in this study. Findings related to research questions (1) and (2) are presented separately in Sections 3.1 and 3.2. The correlation analysis conducted to examine research question (3) is reported in Section 3.3. Section 3.4 further applies a clustering approach to classify EVs based on similarities in their soundscape characteristics. Finally, Sections 4.1 and 4.2 explore potential applications of the results, while Section 4.3 outlines the limitations of the present study.

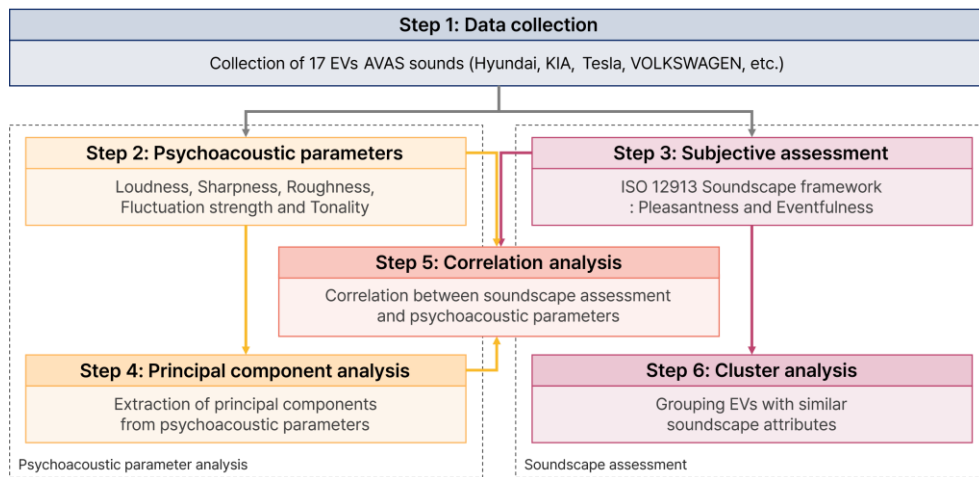


FIG. 1. Framework of this study.

II. METHOD

A. Recording of AVAS sounds

In this study, we collected AVAS sounds from a total of 17 EVs (EV1-17) across 13 different brands, including Hyundai, KIA, Genesis, Tesla, BMW, Volvo, Honda, Chevrolet, Peugeot, and Audi, among others. All AVAS sounds were recorded from outside the vehicle. Three vehicle operating conditions were selected for AVAS sound recording: a stationary condition with drive gear engaged (0 km/h), forward pass-by at 20 km/h, and reverse maneuver

at 6 km/h. However, depending on the regional market specifications of the vehicle, eight EV models (EV10-17) did not emit AVAS sounds under stationary conditions. Therefore, the total number of collected EV AVAS sounds was 43. Details of the measurement environments and settings are illustrated in Fig. 2.

According to ISO 16254²⁶, the recording position was located 2 m in front of the vehicle AVAS. For the stationary condition (D0), AVAS sounds were recorded from Position 1, as shown in Fig. 2(a). Binaural recordings and video capture were made while the vehicle was stationary with the drive gear engaged. Since EV10-17 did not emit AVAS sounds under stationary conditions, they were excluded from the Position 1 recordings. For the forward pass-by condition at 20 km/h (D20) and reverse maneuver at 6 km/h (R6), recordings were conducted from Position 2. In these cases, as shown in Fig. 2(b), the vehicle passed from left to right relative to the microphone at D20 (from the recorder's point of view), whereas at R6 the vehicle reversed from right to left. All recordings were made on a low-noise test track with a surface absorption coefficient of approximately 0.8.

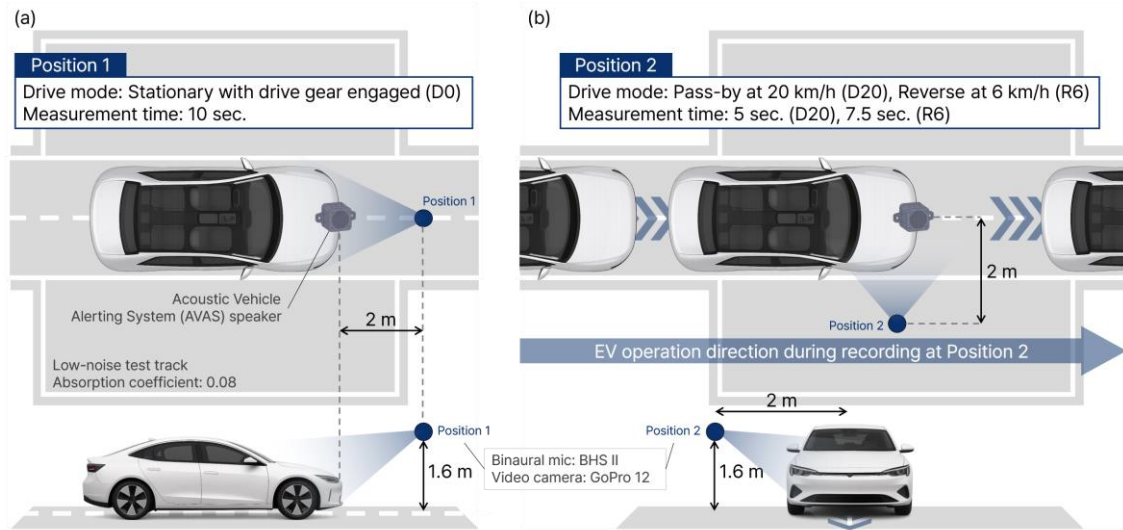


FIG. 2. Audio-visual recording setup for EV AVAS sounds: (a) Position 1 (Stationary condition with drive gear engaged) and (b) Position 2 (Forward pass-by at 20 km/h and reverse maneuver at 6 km/h)

Binaural measurements were carried out at an elevation of 1.60 m, employing a BHS II binaural micro phone (HEAD Acoustics GmbH, Germany) in combination with a SQobold portable sound analyzer (HEAD Acoustics GmbH, Germany). Audio signals were captured with a 24-bit resolution and sampling frequency of 48 kHz to achieve high-quality recordings. Sound pressure levels of each AVAS sound under the three operating conditions were analyzed using ArtemiS software (HEAD Acoustics GmbH, Germany). The A-weighted equivalent sound pressure levels (L_{Aeq}) for each condition are presented in Fig. 3, with values generally ranging within 45 – 67 dB, showing considerable variation across vehicles. In addition, during the binaural recording process, action camera footage (GoPro HERO12, GoPro, USA) was simultaneously captured at both Position 1 and Position 2 (shown in Fig. 2), representing different vehicle movement scenarios.

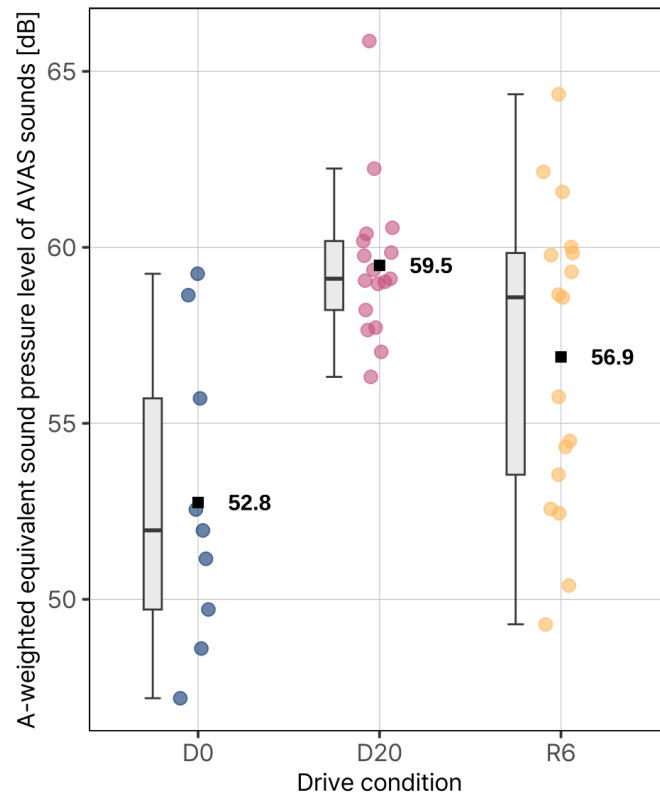


FIG. 3. A-weighted equivalent sound pressure levels of 43 EV AVAS sounds by drive condition (D0, D20, and R6). The black rectangular points represent the mean values.

B. Analysis of psychoacoustic parameters

Five psychoacoustic parameters were analyzed to characterize the acoustic properties of the AVAS sounds: loudness, sharpness, roughness, fluctuation strength, and tonality. All parameters were calculated using an acoustic analysis software package (ArtemiS Suite, HEAD Acoustics GmbH, Germany). Loudness, which represents the perceived intensity of sound by the human auditory system, was computed according to the DIN 45631/A1 standard ¹¹. Sharpness, which reflects the auditory impression of higher frequency components within a sound, was calculated in accordance with the same standard ¹¹. To evaluate temporal modulation in the amplitude of the sounds, both roughness and fluctuation strength were analyzed according to the psychoacoustic model developed by Zwicker and Fastl ¹⁵. Roughness describes the perceptual effect caused by fast, periodic fluctuations in amplitude, typically associated with modulation frequencies between 15 and 300 Hz. Meanwhile, fluctuation strength is related to the perception of slower amplitude variations, usually occurring at modulation frequencies below 20 Hz. Tonality, describing the degree to which tonal components are perceived relative to the background noise, was calculated based on the ECMA-418-2 (2nd Edition) standard ¹². Additionally, mean, maximum, and percentile values (1%, 10%, 50%, and 90%) were calculated.

C. Assessment of perceived affective quality

ISO/TS 12913-2 ²⁴ and 3 ²⁵ provide a structured methodology for its assessment. In this study, to evaluate AVAS sounds from a soundscape perspective, we employed the Method A questionnaire related to the perceived affective quality (PAQ) proposed in ISO/TS 12913-2. The questionnaire consists of eight affective attributes: pleasant, chaotic, vibrant, uneventful, calm, annoying, eventful, and monotonous. These soundscape attributes were rendered into Korean in accordance with the terminology and translation principles proposed in a previously validated reference ²⁹, thereby maintaining both linguistic accuracy and conceptual equivalence.

Participants rated each attribute using a visual analogue scale ranging from 0 (strongly disagree) to 10 (strongly agree). According to Method A outlined in ISO/TS 12913-3, scores for Pleasantness (ISOP) and Eventfulness (ISOE) were calculated using Equations (1) and (2), respectively, based on the PAQ responses.

$$ISOP = \{(p - a) + \cos 45^\circ \cdot (ca - ch) + \cos 45^\circ \cdot (v - m)\} / (4 + \sqrt{32}) \quad (1)$$

$$ISOE = \{(e - u) + \cos 45^\circ \cdot (ca - ch) + \cos 45^\circ \cdot (v - m)\} / (4 + \sqrt{32}) \quad (2)$$

where ' p ' is pleasant, ' a ' is annoying, ' ca ' is calm, ' ch ' is chaotic, ' e ' is eventful, ' u ' is uneventful, ' v ' is vibrant, and ' m ' is monotonous.

Following the standard, the values of ISOP and ISOE were averaged for each AVAS sound across all drive conditions to obtain representative perceptual metrics.

D. Experiment procedure

The listening experiment was conducted in a soundproof chamber, where the background noise level ($L_{Aeq, 3-min}$) was maintained at approximately 25 dB(A) to ensure a controlled acoustic environment. AVAS sounds were presented to participants using open-back headphones (HD650, Sennheiser, Germany), providing high-fidelity reproduction of the binaural recordings.

To enhance participants' immersion in the listening context and simulate realistic AVAS experiences, synchronized visual stimuli were presented. Video recordings were displayed on a television monitor (4K UHD OLED TV, LG, Korea) during the listening session. Only one vehicle was filmed to provide a consistent visual context, while the recorded AVAS sounds from all 17 vehicle models were individually synchronized and presented with the same video. To reduce the influence of possible order-related biases, the playback order of the AVAS sounds stimuli was individually randomized for each participant. All audio-visual stimuli were delivered through a custom-built Unity project (version 2023.2.9f1), which also served as the user interface for participant interaction, and responses were collected directly within the Unity environment.

E. Participants

All experimental procedures complied with established ethical guidelines and received approval from the Institutional Review Board (IRB No. 202402-SB-021-01). Before taking part, participants received written explanations of the study and submitted written informed consent. The study sample consisted of 132 individuals, including 65 males and 67 females, with ages spanning from 19 to 61 years (Mean = 47.43 years and SD = 10.97 years). Detailed demographic information on participants' age and gender is presented in Table 1.

TABLE I. Summary of participant distribution by age group and gender.

Age group	Male (%)	Female (%)	Total (%)
20s	21 (16%)	14 (11%)	35 (27%)
30s	19 (14%)	24 (18%)	43 (32%)
40s	14 (11%)	16 (12%)	30 (23%)
50+	10 (7%)	14 (11%)	24 (18%)
Total	64 (48%)	68 (52%)	132 (100%)

F. Data analysis

To identify the underlying structure of the psychoacoustic characteristics of AVAS sounds, Principal Component Analysis (PCA) was performed using the varimax rotation method. Principal components with eigenvalues greater than 1 were retained. Furthermore, each EV was analyzed separately using one-way repeated-measures analysis of variance (RM ANOVA) tests to examine differences in PAQ arising from variations in AVAS sounds across drive conditions. Sphericity was examined by applying Mauchly's test to the dataset. In cases where the sphericity assumption was not satisfied, adjustments were made using the Greenhouse-Geisser epsilon($\hat{\epsilon}$), in line with the criteria proposed by Blanca et al. (2023). Specifically, the Greenhouse-Geisser correction was adopted for $\hat{\epsilon}$ values below 0.6, whereas the Huynh-Feldt correction was

employed when $\hat{\epsilon}$ ranged from 0.6 to 0.9. Correspondingly adjusted degrees of freedom and p -values for the F -statistics were reported. Post-hoc tests were performed using Bonferroni adjustments to control for multiple comparisons across the 17 vehicles. Effect sizes were quantified using partial eta squared (η_p^2).

In addition, Pearson's correlation coefficients were analyzed to examine the relationships between psychoacoustic parameters and soundscape descriptors. Subsequently, in order to group EVs with similar perceptual characteristics based on ISOP and ISOE, a hierarchical cluster analysis (HCA) was performed using Ward's method with Euclidean distance. To characterize the clusters from the HCA, a PCA was also conducted using ISOP and ISOE values across EVs. All statistical procedures were carried out with the software packages R (version 4.5.0) and jamovi (version 2.3.18).

III. RESULTS

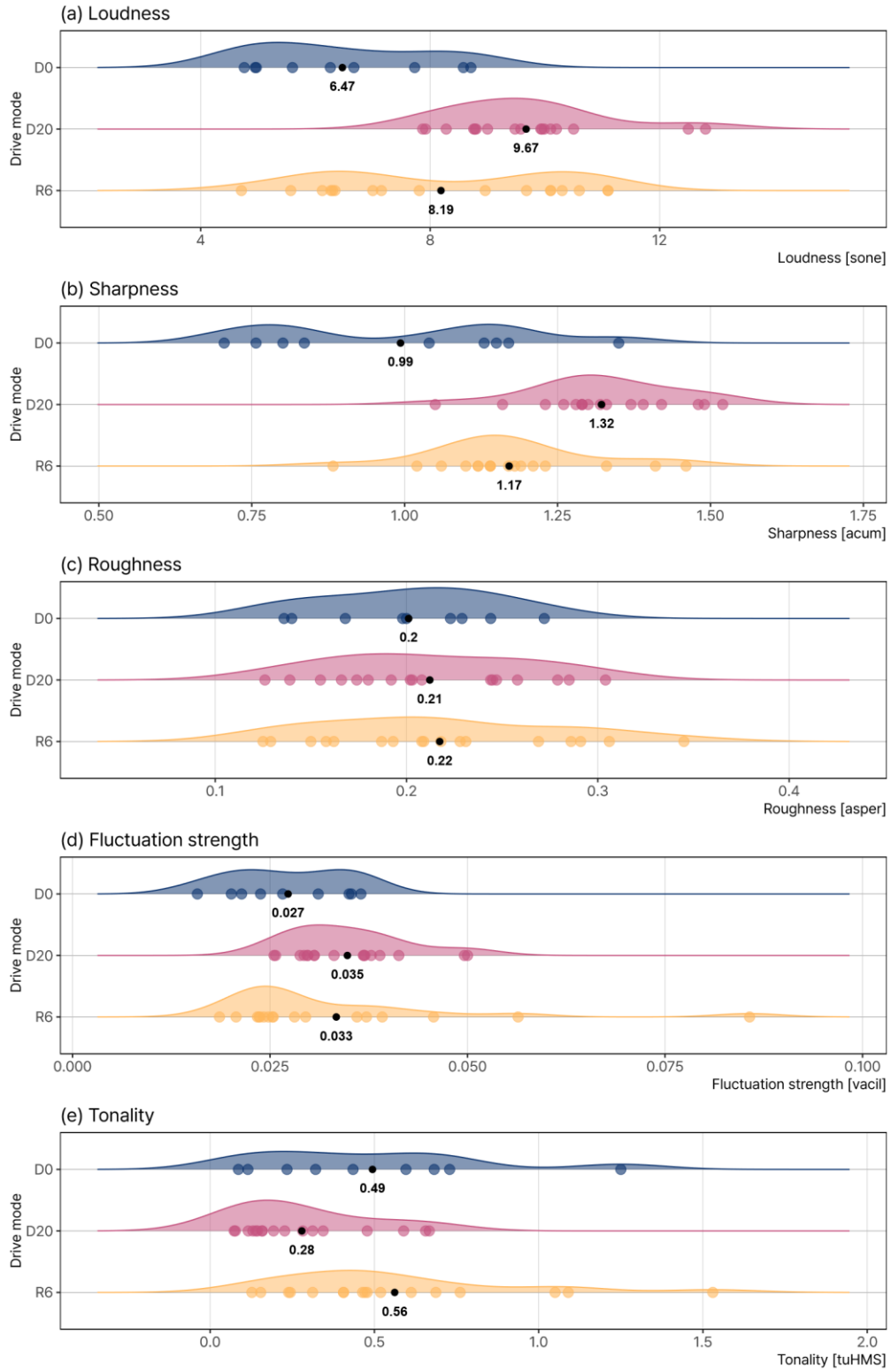
A. Principal components of psychoacoustic parameters of AVAS sounds

To examine the overall psychoacoustic characteristics of AVAS sounds, five key psychoacoustic parameters—loudness, sharpness, roughness, fluctuation strength, and tonality—were computed for 43 AVAS sounds across three drive conditions as shown in Table 2.

Figures 4(a–e) present the distribution of each parameter by drive condition. Regarding loudness, as shown in Fig. 4(a), D0 exhibited the lowest average value of 6.47 sone, followed by 8.19 sone in R6, and the highest value of 9.67 sone in D20. This indicates that AVAS sounds tend to become louder when the vehicle is in motion. Similarly, sharpness, shown in Fig. 4(b), increased with driving speed: D20 had the highest mean (1.32 acum), followed by R6 (1.17 acum), and D0 had the lowest (0.99 acum). Notably, sharpness under D0 exhibited a bimodal distribution, with peaks around 0.75 and 1.2. As shown in Table 2, this indicates that EV2, EV3, EV7, and EV9, which showed values lower than 0.99 acum, inherently contained fewer high-

frequency components compared to other vehicles. In contrast, EV1, EV4, EV5, EV6, and EV8 exhibited values greater than 0.99 acum, with EV1 in particular demonstrating the highest sharpness, indicating a greater presence of high-frequency components. Such differences in loudness and sharpness across driving modes exceed the just noticeable differences (JND) reported in a previous study (You & Jeon, 2008; 0.5 sone for loudness and 0.08 acum for sharpness), suggesting that people would be able to clearly perceive these variations.

Roughness, depicted in Fig. 4(c), remained relatively stable across conditions, with only a 0.01 asper difference in mean values between D0, D20, and R6. In contrast, fluctuation strength, shown in Fig. 4(d), followed a pattern similar to loudness and sharpness, with the lowest average observed in D0 (0.027 vacil), increasing in R6 (0.033 vacil) and peaking in D20 (0.035 vacil). According to a previous study⁵³, the JND values for roughness (0.04 asper) and fluctuation strength (0.012 vacil) indicate that the mean differences observed across driving modes in this study are smaller than the perceptual thresholds and therefore unlikely to be noticeable to listeners. However, the unusually high fluctuation strength observed in R6 was primarily due to EV16, which also exhibited a similar pattern in tonality. Interestingly, tonality, shown in Fig. 4(e), exhibited a trend opposite to the other parameters. The highest mean tonality was observed in R6 (0.56 tu), followed by D0 (0.49 tu), while D20 yielded the lowest average (0.28 tu). These results suggest that AVAS sounds tend to exhibit higher tonality but lower loudness, sharpness, and fluctuation strength when the vehicle is stationary, whereas forward motion reduces tonality while increasing the other parameters.



212

213 FIG. 4. Distribution of psychoacoustic parameters across driving modes: (a) Loudness, (b)

214 Sharpness, (c) Roughness, (d) Fluctuation strength, and (e) Tonality. The black points indicate

215 the mean values of psychoacoustic parameters for each driving mode.

TABLE II. Psychoacoustic parameters of 43 AVAS sounds from 17 EVs by drive condition.

No.	Drive condition	<i>Loudness</i> (sone)	<i>Sharpness</i> (acum)	<i>Roughness</i> (asper)	<i>Fluctuation</i> (vacil)	<i>Tonality</i> (tu)
EV1	D0	8.8	1.35	0.17	0.037	1.25
	D20	13.7	1.39	0.14	0.026	0.34
	R6	6.4	1.14	0.13	0.02	0.76
EV2	D0	10.4	0.80	0.23	0.027	0.68
	D20	18.4	1.33	0.20	0.037	0.13
	R6	11.0	1.14	0.21	0.021	0.24
EV3	D0	5.4	0.76	0.20	0.016	0.23
	D20	15.0	1.16	0.21	0.033	0.28
	R6	13.4	0.88	0.29	0.024	0.41
EV4	D0	7.2	1.17	0.14	0.031	0.60
	D20	13.6	1.52	0.13	0.029	0.48
	R6	14.4	1.19	0.19	0.037	1.09
EV5	D0	5.5	1.13	0.14	0.021	0.32
	D20	16.0	1.28	0.28	0.038	0.31
	R6	15.3	1.06	0.35	0.028	0.41
EV6	D0	7.2	1.04	0.20	0.035	0.44
	D20	14.1	1.42	0.18	0.039	0.67
	R6	8.3	1.21	0.13	0.025	0.69
EV7	D0	5.2	0.71	0.22	0.020	0.09
	D20	15.8	1.48	0.16	0.037	0.16
	R6	12.9	1.46	0.16	0.024	0.13
EV8	D0	9.4	1.15	0.27	0.024	0.12

No.	Drive condition	<i>Loudness</i> (sone)	<i>Sharpness</i> (acum)	<i>Roughness</i> (asper)	<i>Fluctuation</i> (vacil)	<i>Tonality</i> (tu)
EV9	D20	13.8	1.26	0.25	0.041	0.077
	R6	12.3	1.10	0.31	0.030	0.15
	D0	6.2	0.84	0.24	0.035	0.73
	D20	12.4	1.29	0.19	0.030	0.12
	R6	9.0	1.12	0.23	0.024	0.47
EV10	D20	13.0	1.30	0.17	0.031	0.66
	R6	7.1	1.12	0.16	0.019	0.46
EV11	D20	14.7	1.49	0.29	0.037	0.14
	R6	10.2	1.41	0.29	0.025	0.31
EV12	D20	17.2	1.23	0.30	0.030	0.14
	R6	13.1	1.33	0.23	0.056	0.48
EV13	D20	12.8	1.29	0.20	0.026	0.16
	R6	10.2	1.02	0.21	0.025	0.52
EV14	D20	15.6	1.05	0.26	0.05	0.59
	R6	14.4	1.23	0.27	0.036	1.05
EV15	D20	11.6	1.29	0.25	0.029	0.23
	R6	8.5	1.14	0.22	0.046	0.61
EV16	D20	20.4	1.32	0.24	0.050	0.07
	R6	18.1	1.18	0.15	0.086	1.53
EV17	D20	14.3	1.37	0.17	0.031	0.19
	R6	8.7	1.17	0.19	0.039	0.24

A PCA was applied to explore the latent structure of psychoacoustic characteristics among the 43 AVAS sounds. The analysis was based on a matrix comprising 43 sound samples and 25 psychoacoustic variables, with varimax rotation employed to obtain orthogonal components.

The Kaiser-Meyer-Olkin index yielded a value of 0.61, and Bartlett's test of sphericity was statistically significant ($\chi^2(300) = 2163, p < 0.001$), confirming the suitability of the data for PCA.

As shown in Table 3, five principal components with eigenvalues greater than 1 were extracted, accounting for a cumulative 87.6% of the total variance in the dataset. The individual contributions to variance were 22.0% (PC1), 21.9% (PC2), 18.1% (PC3), 17.7% (PC4), and 8.0% (PC5). Each component represents a distinct psychoacoustic dimension of AVAS sounds. PC1 showed high positive loadings on sharpness-related features and reflects the perceived spectral brightness or sharpness of the sound. PC2 was primarily associated with tonality-related indicators, including the 50th and 90th percentile values, and can be interpreted as representing the clarity of the sound. PC3 was defined by roughness-related parameters, representing the textural qualities of the sound. PC4 loaded strongly on fluctuation strength and loudness features, capturing the overall intensity and temporal modulation of the AVAS signal. Lastly, PC5 was largely characterized by the 90th percentile of loudness, likely reflecting background amplitude peaks and serving as a supplementary dimension. These results reflect the multidimensional psychoacoustic characteristics of AVAS sounds, encompassing spectral brightness (PC1), pitch clarity (PC2), textural roughness (PC3), modulation intensity (PC4), and background loudness (PC5).

239 TABLE III. Principal component loadings for psychoacoustic parameters of AVAS sounds.

Psychoacoustic parameters	Principal components (Explained Variance, %)				
	1 (22.0%)	2 (21.9%)	3 (18.1%)	4 (17.7%)	5 (8.0 %)
S_{Mean}	0.97	-0.05	-0.14	0.04	0.01
S_{50}	0.96	-0.06	-0.11	0.08	-0.01
S_{10}	0.95	-0.09	-0.07	0.10	-0.05
S_{Max}	0.85	-0.05	-0.17	0.07	-0.03
S_{90}	0.72	0.13	-0.25	-0.08	0.24
T_{Mean}	-0.08	0.94	-0.20	0.06	-0.01
T_{50}	-0.11	0.91	-0.17	0.09	0.07
T_{10}	0.00	0.85	-0.25	0.13	-0.21
F_{90}	-0.16	0.82	0.04	-0.25	0.11
T_{Max}	0.02	0.81	-0.18	0.13	-0.30
T_{90}	-0.17	0.75	-0.01	-0.35	0.36
F_{50}	0.27	0.68	-0.03	0.38	-0.18
R_{Mean}	-0.11	-0.12	0.96	0.17	0.05
R_{10}	-0.11	-0.12	0.96	0.18	0.05
R_{Max}	-0.09	-0.07	0.95	0.20	-0.01
R_{50}	-0.27	-0.30	0.80	0.14	0.34
R_{90}	-0.30	-0.36	0.69	0.10	0.44
F_{10}	-0.16	0.06	0.19	0.90	0.08
F_{Max}	-0.30	0.12	0.16	0.80	0.09
N_{Max}	0.55	-0.00	0.20	0.75	0.08
N_{10}	0.53	-0.11	0.22	0.73	0.21
N_{Mean}	0.44	-0.06	0.27	0.68	0.44
F_{Mean}	0.24	0.54	0.01	0.67	-0.10
N_{50}	0.40	-0.19	0.31	0.59	0.54
N_{90}	0.11	-0.04	0.20	0.30	0.86

B. Soundscape assessment of AVAS sounds

In this study, AVAS sounds were assessed using the ISO 12913 perceived affective quality attributes, focusing specifically on ISOP and ISOE. The computed ISOP and ISOE scores for 17 EVs (EV1–EV17) are presented in Fig. 5 and Fig. 6, respectively. Differences in ISOP and ISOE across drive conditions were examined using a one-way repeated-measures ANOVA with Bonferroni-adjusted post-hoc tests, with the corresponding results provided in Appendix A. With respect to pleasantness, Figure 5 illustrates that EV6 achieved the highest average ISOP value (0.31), while EV16 showed the lowest (−0.14). For EV2, EV7, and EV8, the ISOP scores under D0 condition were negative, indicating that the AVAS sounds in this condition were perceived as unpleasant. For most EVs—except EV3, EV5, and EV6—the ISOP scores under D0 condition were lower than those observed in the other drive conditions. Significant differences were found in EV1 (D20: $t = -5.16, p < 0.001$; R6: $t = -5.64, p < 0.001$), EV4 (D20: $t = -3.87, p < 0.001$; R6: $t = -2.67, p = 0.03$), EV7 (D20: $t = -4.11, p < 0.001$; R6: $t = -7.57, p < 0.001$), and EV8 (D20: $t = -6.64, p < 0.001$). In contrast, the D20 condition resulted in higher ISOP scores than R6 for many EVs, such as EV2 ($t = 3.06, p = 0.01$), EV5 ($t = 2.52, p = 0.04$), EV8 ($t = 4.96, p < 0.001$), EV13 ($t = 2.99, p = 0.00$), EV14 ($t = 4.56, p < 0.001$), EV16 ($t = 2.20, p = 0.03$), EV17 ($t = 3.33, p = 0.01$). Although EV3, EV4, and EV15 also showed higher ISOP scores in D20 than in R6, the differences were not statistically significant. Conversely, only EV7 ($t = -3.76, p < 0.001$), EV9 ($t = -2.70, p = 0.02$), and EV12 ($t = -3.05, p = 0.00$) exhibited higher ISOP scores in R6 compared to D20. These findings suggest that AVAS sounds under the stationary condition (D0) were generally perceived as less pleasant, whereas pass-by at 20 km/h (D20) tended to be perceived as more pleasant than reverse at 6 km/h (R6).

In terms of eventfulness, as shown in Fig. 6, EV12 and EV16 exhibited the highest ISOE values (0.30), while EV3 and EV13 showed the lowest (−0.01). Under the D0 condition, all EVs except EV3 yielded positive ISOE scores, suggesting that stationary AVAS sounds are generally

perceived as eventful. In contrast, ISOE scores under R6 were generally lower than those under both D0 and D20. All EVs showed lower ISOE values in R6 compared to D0, with significant differences observed in all cases except EV3, EV5, and EV8 (EV1 ($t = 9.39, p < 0.001$), EV2 ($t = 4.39, p < 0.001$), EV4 ($t = 4.63, p < 0.001$), EV6 ($t = 6.87, p < 0.001$), EV7 ($t = 9.27, p < 0.001$), EV9 ($t = 6.48, p < 0.001$)). Similarly, when compared to D20, 14 out of 17 EVs showed lower ISOE scores under R6, and significant differences were found in 11 EVs (EV1 ($t = 9.83, p < 0.001$), EV2 ($t = 4.35, p < 0.001$), EV3 ($t = 6.92, p < 0.001$), EV5 ($t = 6.16, p < 0.001$), EV6 ($t = 7.69, p < 0.001$), EV7 ($t = 7.08, p < 0.001$), EV9 ($t = 5.50, p < 0.001$), EV10 ($t = 9.21, p < 0.001$), EV11 ($t = 6.41, p < 0.001$), EV13 ($t = 2.57, p = 0.01$), EV15 ($t = 2.46, p = 0.02$)) based on Bonferroni-corrected post hoc tests. The only exception was EV12 ($t = 9.83, p < 0.001$), which exhibited significantly higher ISOE in R6 than in D20. In summary, AVAS sounds tended to show higher ISOE values under stationary conditions and lower ISOE values under reverse conditions. Overall, these results suggest that stationary AVAS sounds were generally perceived as less pleasant but highly eventful, whereas AVAS sounds during forward driving were perceived as more pleasant than during reverse driving, with reverse AVAS sounds being more eventful than forward ones. The distributions of ISOP and ISOE scores across all EVs are presented in Fig. 7. The overall distribution of ISOP and ISOE assessments by EV was centralized, and the distribution across EVs was broadly similar. Notably, ISOP and ISOE scores were not significantly correlated ($r = -0.20, p = 0.20$), indicating that pleasantness and eventfulness were perceived as independent dimensions regarding AVAS sounds.

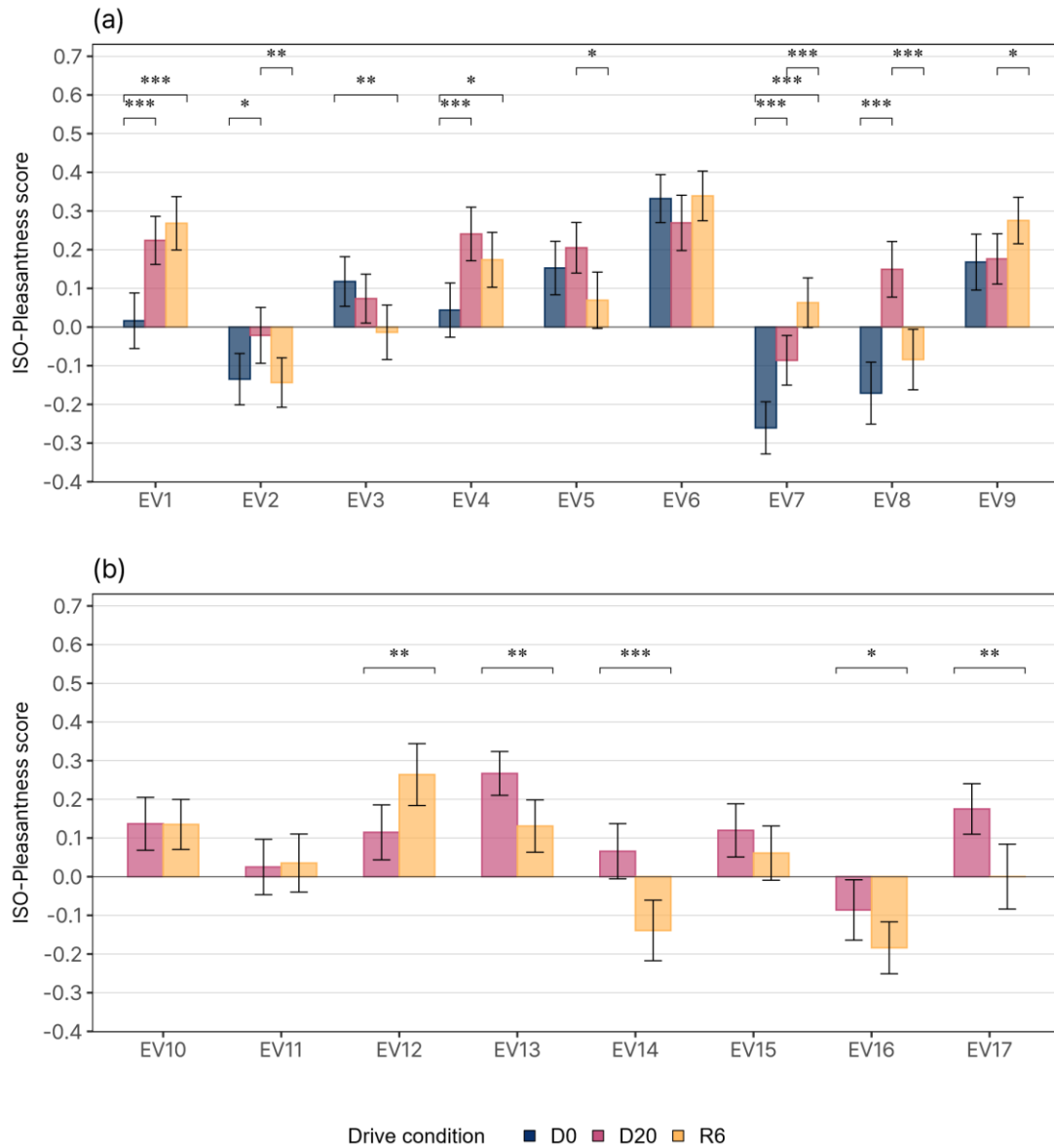


FIG. 5. Mean ISOP scores of EVs across three drive conditions. Error bars indicate the confidence interval. Panel (a) shows EV1-EV9 with AVAS sounds in the D0 condition, and panel (b) shows EV10-EV17 without AVAS sounds in the D0 condition. Asterisks denote statistically significant differences identified through repeated-measures ANOVA followed by Bonferroni-corrected post-hoc test ($p < 0.05$, $p < 0.01$, and $p < 0.001$).

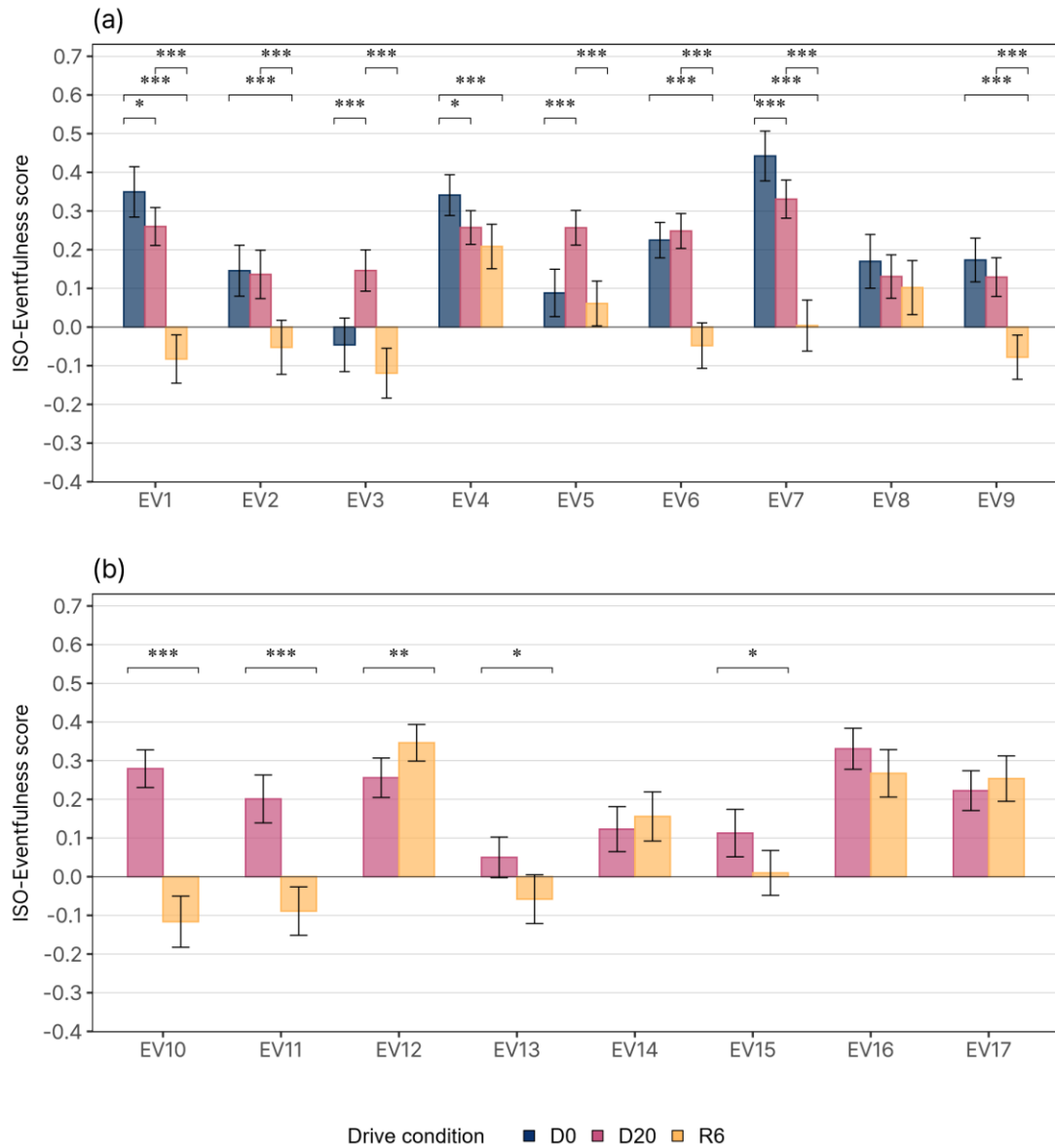
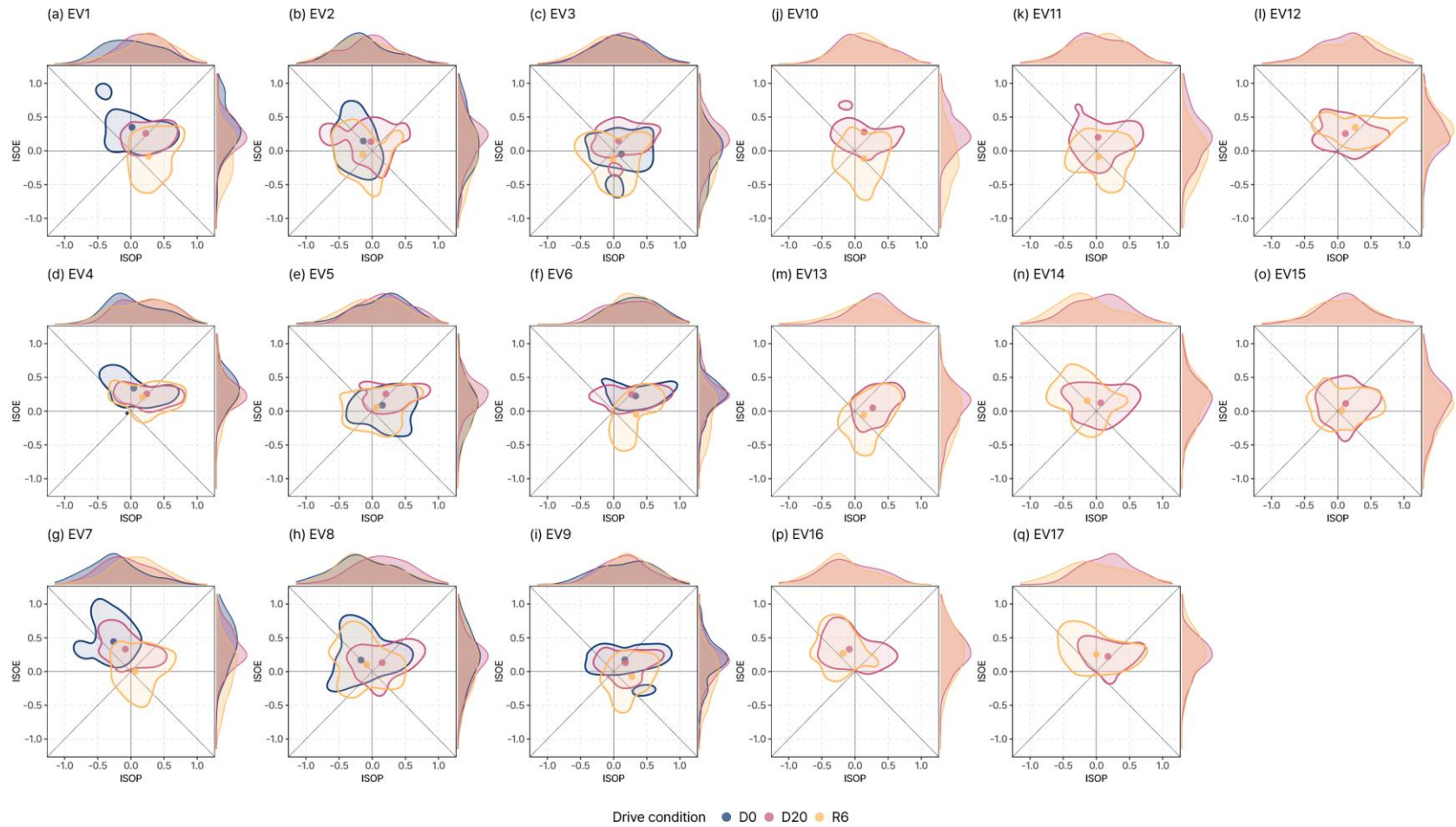


FIG. 6. Mean ISOE scores of EV across three drive conditions. Error bars indicate the 95% confidence interval. Panel (a) shows EV1-EV9 with AVAS sounds in the D0 condition, and panel (b) shows EV10-EV17 without AVAS sounds in the D0 condition. Asterisks denote statistically significant differences identified through repeated-measures ANOVA followed by Bonferroni-corrected post-hoc test (* $p < 0.05$, ** $p < 0.01$, and *** $p < 0.001$).



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300 FIG. 7. Means and contours indicating 50% probability regions derived from kernel density estimation of ISOP and ISOE scores for each

301 EVs.

C. Correlation between psychoacoustic parameters and soundscape attributes

Pearson's correlation analysis was conducted to explore the relationships between the psychoacoustic parameters of EV AVAS sounds and the soundscape attributes, as presented in Table 4. The analysis included five representative psychoacoustic indicators derived from PCA, together with L_{Aeq} . As AVAS sounds vary depending on the drive conditions, ISOP and ISOE scores were averaged separately for D0, D20, and R6. Likewise, L_{Aeq} and psychoacoustic parameters were averaged by drive condition. When ISOP and ISOE were analyzed as overall means, L_{Aeq} and psychoacoustic parameters were averaged accordingly and included in the correlation analysis. For ISOP, the correlation analysis revealed significant negative relationships with F_{10} ($r = -0.41, p = 0.01$) and N_{90} ($r = -0.54, p < 0.001$). These associations were more pronounced under the D20 condition ($F_{10}: r = -0.63, p = 0.01$; $N_{90}: r = -0.51, p = 0.02$) and the R6 condition ($F_{10}: r = -0.60, p = 0.01$; $N_{90}: r = -0.76, p < 0.001$), whereas no significant correlations were found under D0. These results suggest that higher values of F_{10} and N_{90} are associated with lower perceived pleasantness during EV driving.

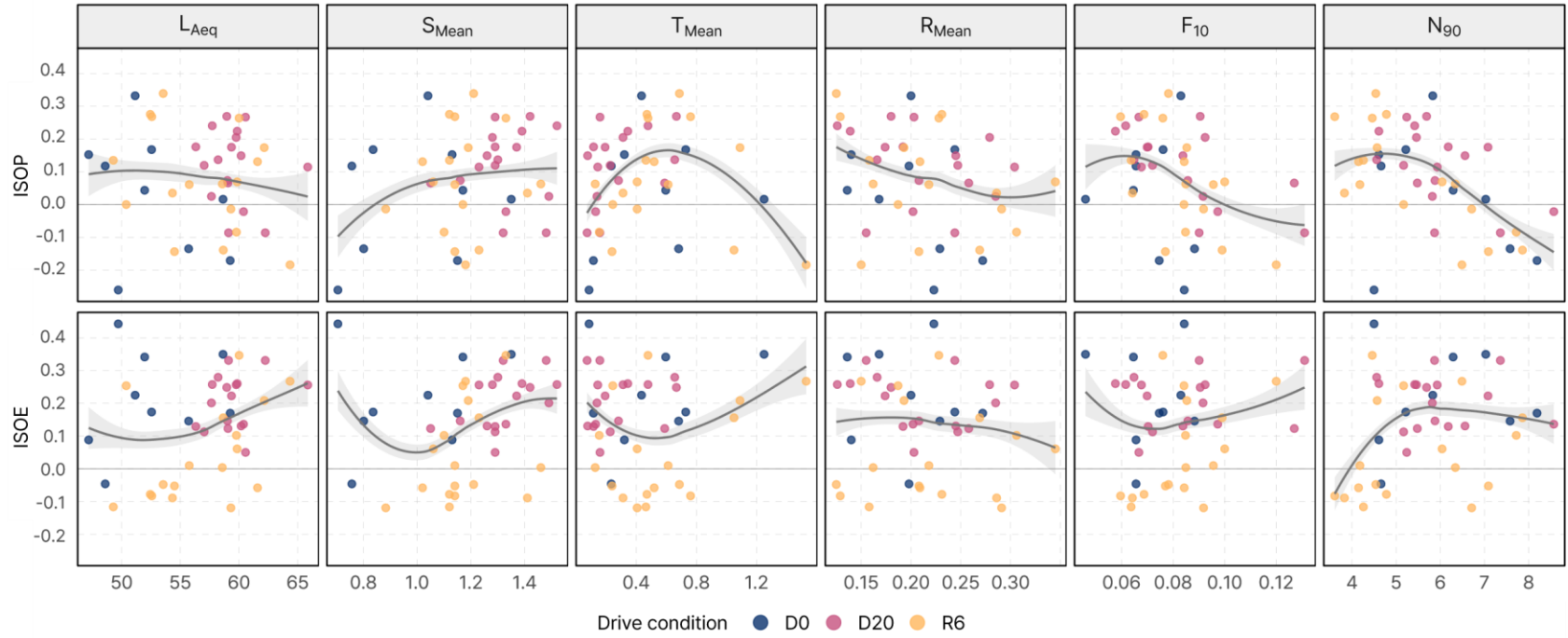
ISOE demonstrated a positive and significant correlation with S_{Mean} ($r = 0.48, p = 0.05$) in the D20 condition, suggesting that increased sharpness during forward driving contributed to the perception of eventfulness. Under the R6 condition, F_{10} ($r = 0.50, p = 0.04$) was positively correlated with ISOE, indicating that more modulated AVAS sounds were perceived as more eventful during reverse driving. Apart from these cases, ISOE did not exhibit significant correlations with the physical or psychoacoustic indicators, suggesting that representative physical and psychoacoustic attributes of AVAS sounds are generally not strongly associated with eventfulness. Overall, these findings indicate that the relationships between physical indicators and soundscape attributes are dependent on drive condition. Loudness and fluctuation strength emerged as the most influential psychoacoustic parameters, showing significant associations with pleasantness. However, under the D0 condition, no significant correlations

were observed, suggesting the absence of clear relationships between the physical indicators and soundscapes attributes in the stationary state. Meanwhile, eventfulness showed significant associations with sharpness and fluctuation strength only in a few cases, while most correlations were non-significant.

TABLE IV. Pearson's correlation coefficients between soundscape attributes and psychoacoustic parameters. (* $p < 0.05$, ** $p < 0.01$, and *** $p < 0.001$)

	Drive condition	L_{Aeq}	S_{Mean}	T_{Mean}	R_{Mean}	F_{10}	N_{90}
ISOP	Mean	-0.12	0.24	0.02	-0.30	-0.41**	-0.54***
	D0	-0.40	0.21	0.23	-0.40	-0.20	-0.35
	D20	-0.18	0.04	0.40	-0.26	-0.63**	-0.51*
	R6	-0.32	0.09	-0.09	-0.34	-0.60*	-0.76***
ISOE	Mean	0.26	0.23	0.08	-0.10	0.12	0.23
	D0	0.25	0.23	0.24	-0.11	-0.03	0.11
	D20	0.24	0.48*	0.18	-0.21	0.09	-0.06
	R6	0.46	0.28	0.38	0.01	0.50*	0.21

Nevertheless, the lack of significant correlations between perceptual responses and other indicators which are key parameters when design AVAS sounds, such as sharpness and tonality, necessitates further examination. Figure 8 presents the observed ISOP and ISOE scores as a function of the six representative psychoacoustic indicators and L_{Aeq} . In each panel, observed data points are shown together with loess-based smoothed curves and their 95% confidence interval. The observed data points generally exhibited an approximately circular distribution, consistent with the low linear correlation coefficients obtained in Pearson's correlation analysis. However, the loess-based smoothed curves revealed nonlinear tendencies, which were particularly pronounced in the relationships between ISOP/ISOE and S_{Mean} or T_{Mean} . These findings suggest that the relationships between ISOP/ISOE and the physical and psychoacoustic indicators cannot be fully captured by simple linear correlations and may instead involve more complex nonlinear structures.



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FIG. 8. Scatter plots illustrating the relationships between ISOP and ISOE scores and L_{Aeq} and five representative psychoacoustic indicators (S_{Mean} , T_{Mean} , R_{Mean} , F_{10} , and N_{90}) across different drive conditions (D0, D20, and R6). Each point represents the averaged value for an individual EV. Solid lines indicate loess-smoothed trends with shaded areas representing the 95% confidence intervals.

D. Clustering EVs based on soundscape attributes of AVAS sounds

The AVAS sounds of individual EVs differed substantially, employing strategies such as frequency shifting, loudness adjustments, or the addition of warning tones to maximize audibility at low speeds ³⁴. Accordingly, the ISO soundscape scores—ISOP and ISOE—of EV AVAS sounds exhibited distinct characteristics across drive conditions. Furthermore, as discussed in Section 4.2, consistent patterns were observed in the differences in ISOP and ISOE between drive conditions, with certain EVs showing similar response trends. To examine these similarities, hierarchical cluster analysis (HCA) was conducted to group the 17 EVs based on their ISOP and ISOE profiles, with the aim of identifying cluster-specific features.

For the cluster analysis, both the ISOP and ISOE scores for each drive condition and the inter-condition differences were included as variables, ensuring that variations across drive conditions also contributed to the clustering criteria. The D0 condition was excluded because only 9 of the 17 vehicles emitted AVAS sounds. As a result, six variables derived from the D20 and R6 conditions were used: $ISOP_{D20}$, $ISOE_{D20}$, $ISOP_{R6}$, $ISOE_{R6}$, and the inter-condition differences between D20 and R6 ($ISOP_d = ISOP_{D20} - ISOP_{R6}$; $ISOE_d = ISOE_{D20} - ISOE_{R6}$). Positive difference values indicate higher scores in D20 compared to R6. Based on these variables, hierarchical clustering using Ward's method and Euclidean distance was performed. As shown in Fig. 9, three distinct clusters were identified: Cluster 1 (EV1, EV6, EV9, and EV10), Cluster 2 (EV2, EV3, EV5, EV7, EV11, EV13, and EV15), and Cluster 3 (EV4, EV8, EV12, EV14, EV16, and EV17).

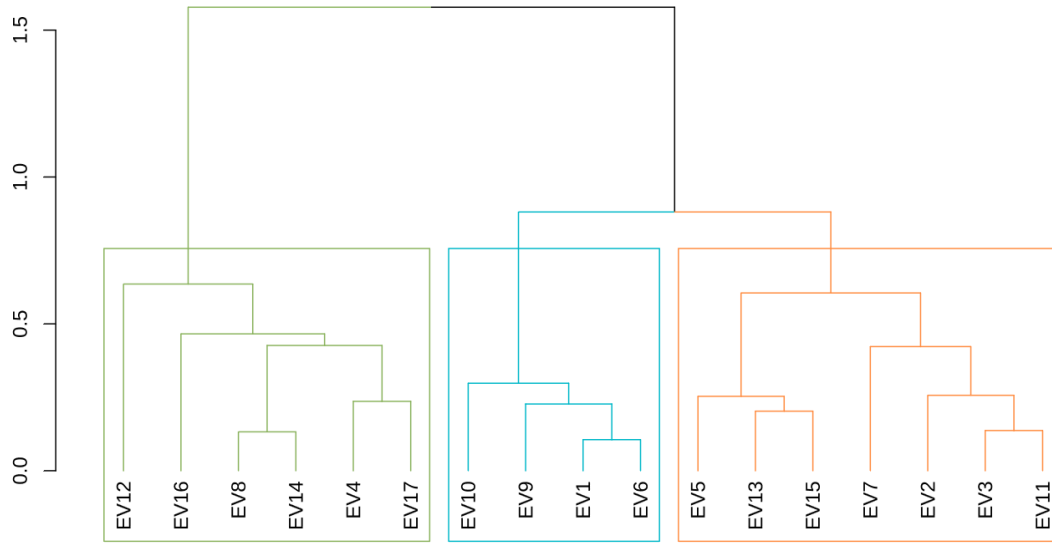


FIG. 9. Results of hierarchical clustering dendrogram. Blue indicates Cluster 1, orange indicates Cluster 2, and green indicates Cluster 3.

Summarizing the acoustic characteristics of each cluster, Cluster 1 comprised EVs that were most distinct from traditional ICE vehicles, featuring uniquely EV-specific sounds. Cluster 2 included EVs that were comparatively similar to ICE vehicles, such as EV2 and EV7, which produced engine-like sounds. Cluster 3 was characterized by the addition of repetitive warning tones during reverse driving in all EVs except EV8; this feature was otherwise absent in other clusters, except for EV15 in Cluster 2. Figure 10 illustrates the average values of the clustering variables for each group. Cluster 1 is characterized by high pleasantness (ISOP) across all drive conditions, as well as higher ISOE in D20 but lower ISOE in R6 (i.e., high $ISOE_d$). Cluster 2 exhibits a similar eventfulness pattern (high $ISOE_d$) to Cluster 1 but with overall lower pleasantness. Cluster 3 shows a pleasantness pattern comparable to Cluster 2 but is distinguished by consistently high eventfulness across drive conditions. In summary, Cluster 1 differs from Clusters 2 and 3 by maintaining high ISOP regardless of drive condition, whereas Cluster 3 differs from Clusters 1 and 2 by maintaining

high ISOE across all conditions. Thus, Cluster 2 may be considered a neutral group, sharing characteristics of both Cluster 1 and Cluster 3.

To further examine how psychoacoustic parameters differed across clusters, the mean values of the five parameters under each drive condition were calculated for each cluster and are presented in Fig. 11. Loudness (Fig. 11(a)) and fluctuation strength (Fig. 11(d)) showed a progressive increase from Cluster 1 to Cluster 3. Sharpness (Fig. 11(b)) remained relatively consistent across clusters, with cluster 2 consistently exhibiting intermediate values between Clusters 1 and 3. Roughness (Fig. 11(c)) was slightly lower in Cluster 1 compared to the other groups, whereas tonality (Fig. 11(e)) was lowest in Cluster 2. In summary, Cluster 1, which was evaluated as the most pleasant, was associated with the lowest average loudness, fluctuation strength, and roughness. Cluster 2, which was characterized by neutral soundscape attributes, exhibited the lowest average tonality. Cluster 3, which was evaluated as the most eventful, was distinguished by the highest loudness and fluctuation strength.

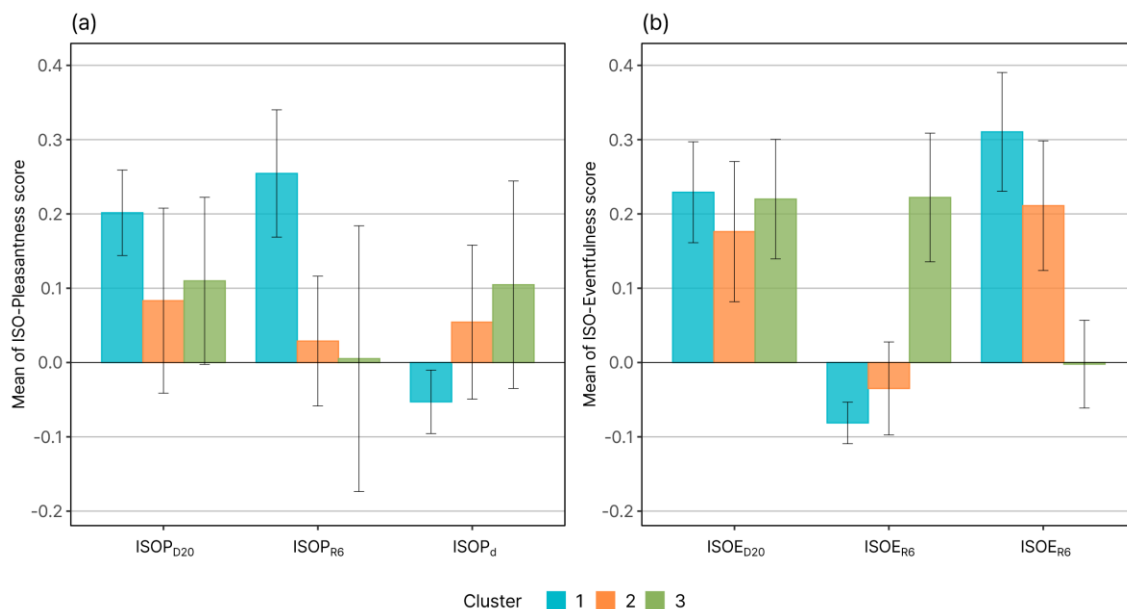
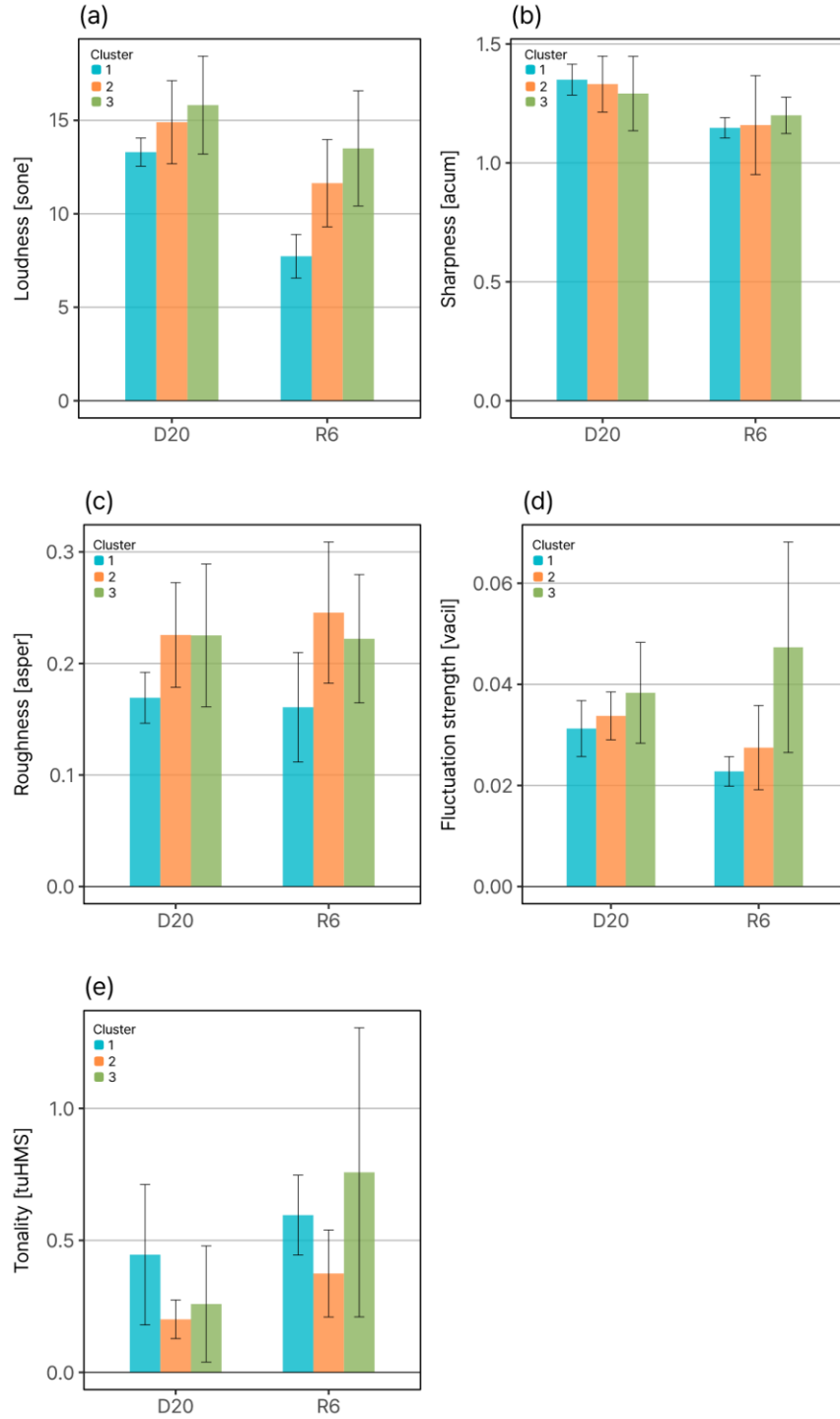


FIG. 10. Cluster-wise mean values of variables used in hierarchical clustering: (a) ISOP scores and (b) ISOE scores. Error bars represent standard deviations.



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400 FIG. 11. Mean psychoacoustic parameters across drive conditions for each cluster: (a) Loudness, (b)

401 Sharpness, (c) Roughness, (d) Fluctuation strength, and (e) Tonality. Error bars represent standard

402 deviations.

IV. DISCUSSION

A. Psychoacoustic parameters as indicators for perceptions of AVAS sounds

The psychoacoustic parameters are regarded as key metrics for explaining pedestrian detectability and perceptual salience of AVAS sounds in design^{6,45}. Accordingly, this study examined the correlations between soundscape attributes (ISOP and ISOE) and the representative psychoacoustic parameters of AVAS sounds extracted through PCA across different drive conditions. The results showed that ISOP and ISOE were not significantly correlated with most psychoacoustic parameters. Only F_{10} and N_{90} were significantly correlated with ISOP, which are associated with PC 4 and PC5, respectively.

While prior studies^{3,13,19} have emphasized the importance of incorporating psychoacoustic features for a comprehensive evaluation of soundscapes—and have frequently reported significant correlations between such parameters and soundscape appraisals—such linear relationships were not clearly observed in the present study. Instead, in this study, nonlinear tendencies were observed in soundscape evaluation as psychoacoustic parameters changed, as shown in Fig. 8. This finding suggests that the relationship between psychoacoustic characteristics of AVAS sounds and soundscape evaluations cannot be sufficiently explained by simple linear correlations. This implies that future studies should consider nonlinear relationships between psychoacoustic parameters and perceptions of AVAS sounds.

Given these nonlinear tendencies, it is worth considering how psychoacoustic parameters are currently utilized in AVAS sound design. When designing AVAS sounds, strategies often prioritize noticeability by increasing loudness⁶, employing high-frequency warning sounds to enhance sharpness²², or maintaining a certain level of tonality³³. Within this context, psychoacoustic parameters may still serve as potential indicators for predicting soundscape evaluations in a nonlinear manner. Importantly, previous studies^{6,22,33} have commonly included annoyance

assessment alongside detectability measures, aiming to identify optimal psychoacoustic ranges that ensure sufficient detection performance while preventing excessive increases in annoyance. A similar perspective may be necessary for soundscape-based evaluations of AVAS sounds, highlighting the need to balance perceptual salience with overall acoustic comfort.

To better account for observed nonlinear tendencies, it is also necessary to examine whether the soundscape attributes used as dependent variables adequately capture the perceptual responses to AVAS sounds. Current soundscape models (Pleasantness-Eventfulness) are designed to evaluate entire acoustic environments and account for the holistic perception of sound in context^{2,32}. By contrast, this study focused on AVAS sounds in isolation. As a result, the applicability of the ISO soundscape model to the evaluation of AVAS sounds may be inherently limited, and the derived ISOP and ISOE scores may not sufficiently reflect trends driven by their psychoacoustic characteristics. As shown in Fig. 7, regarding the ISOP and ISOE scores, 17 EVs were limited in their ability to discriminate the perceptual characteristics of AVAS sounds, exhibiting a similar spherical distribution. This suggests that the model may lack the discriminatory power needed to differentiate among various AVAS sound characteristics. In other words, the standardized set of eight perceptual attributes defined in ISO 12913 may not adequately capture the nuanced affective responses of AVAS sounds. Accordingly, to better examine the relationships between physical indicators and the perceived affective quality of AVAS sounds, alternative perceptual attributes or adapted evaluative models may be required.

B. Potential influence of EV AVAS sound on urban soundscapes

In urban soundscapes, traffic noise constitutes one of the most dominant acoustic elements. From a soundscape perspective, however, such traffic noise is treated as an undesirable source^{21,27,28,38,51,52}. Consequently, several urban soundscape design strategies, such as introducing pleasant natural sounds^{10,55}, planting street trees³⁷, and incorporating green walls⁵, have been proposed to

minimize the negative impacts of traffic noise. However, with the transition from ICE vehicles to EVs, the primary sources of traffic noise are also undergoing fundamental changes. This transition alters the spectral characteristics of traffic noise, and reductions in noise levels at low vehicle speeds have been reported⁴. Some studies indicate that the urban acoustic environment may improve with increased EV adoption^{4,8}, whereas others argue that these changes could potentially worsen overall soundscape quality in urban environments³⁰.

To date, AVAS research has largely focused on cognitive differences related to frequency content or sound levels^{40,48}, affective evaluations that inform vehicle sound design^{45,46}, and pedestrian detectability¹⁷. These efforts have led to a range of AVAS sound design strategies^{42,49}. Recent research has shown that EV sound designs incorporating more progressive and futuristic concepts are perceived by listeners as more pleasant and emotionally engaging³⁵. A similar trend was reflected in the clustering results of the present study, where EVs in Cluster 1—whose AVAS sounds were most dissimilar to conventional engine noise—showed the highest average ISOP scores.

To compare the traffic noise produced by ICE and AVAS using the Pleasantness-Eventfulness model, Figure 12 presents the soundscape evaluation results for EVs in Cluster 1 and soundscape data from location S5, as reported in a previous study by Hong and Jeon²¹, where ICE traffic noise was perceived as the most dominant sound. As illustrated in Fig. 12, the soundscape of the S5 was characterized by a distribution skewed toward negative pleasantness scores, reflecting conventional perceptions of traffic-dominated urban environments. In contrast, the results for Cluster 1 EVs demonstrate a clear shift toward positive ISOP pleasantness scores, regardless of drive condition. This contrast suggests a perceptual difference from traditional traffic noise evaluations, indicating that sounds produced by AVAS-equipped electric vehicles may not align with the negative associations generally linked to traffic noise. These findings also imply that, as EVs increasingly

dominate urban traffic, perceptual tendencies associated with the traffic noise category itself may shift.

Conversely, EVs in Cluster 3, which incorporated additional repetitive warning tones during reverse driving at 6 km/h, resulting in higher ISOE scores. This suggests that repetitive warning tones may increase the eventfulness of urban soundscape. However, it remains unclear whether such increases in eventfulness would ultimately exacerbate or alleviate annoyance in real-world urban environments. Therefore, future AVAS sound designs should extend beyond considerations of pedestrian detectability and brand identity alone, and instead account for the broader implications for urban soundscapes, including the potential to contribute to improved acoustic comfort and perceptual quality in urban environments.

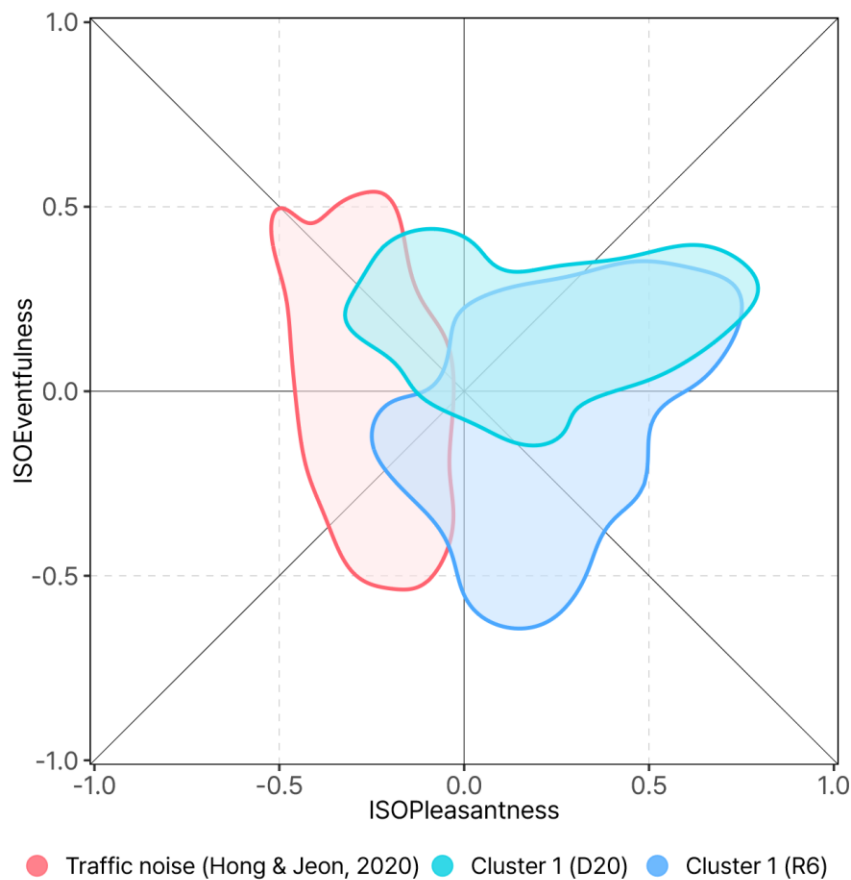


FIG. 12. Comparison of soundscape evaluation distributions in the ISO Pleasantness-Eventfulness model between traffic noise-dominated urban soundscapes reported by Hong & Jeon (2020) and AVAS soundscapes of EVs in Cluster 1, which consists exclusively of EVs whose AVAS sounds are acoustically dissimilar to ICE vehicle noise, under D20 and R6 drive conditions. (The D0 condition is not shown, as it was excluded from the clustering analysis.) Shaded contours indicate the 50% probability regions derived from kernel density estimation.

C. Limitations and future study

To capture perceptual differences across driving contexts, AVAS sounds were collected under various drive conditions (stationary (0 km/h), forward driving (20 km/h), and reverse driving (6 km/h)). However, due to the need for high-quality recordings, data collection was restricted to a limited stretch of low-noise pavement. As a result, the length of the recordings varied across conditions, potentially limiting the comparability of some cases. Future studies should aim to improve standardization in measurement environments and recording procedures to ensure more consistent data.

In addition, the sample was limited to 17 EV models, which may not fully represent the diversity of the current EV market and may not have yielded statistically significant results. Future research should include a broader range of manufacturers and vehicle types. Furthermore, field studies in actual urban soundscapes are needed to comprehensively examine how AVAS sounds influence the overall urban acoustic environment.

V. CONCLUSION

This study examined both physical and perceptual characteristics of AVAS sounds from 17 EV models, combining psychoacoustic analyses with a soundscape-based evaluation framework. A total of 25 psychoacoustic variables were extracted and summarized using PCA resulting in five principal

components that captured 87.6% of the total variance. These components represented key perceptual dimensions of AVAS sounds: spectral brightness (PC1), pitch clarity (PC2), modulation intensity (PC3), textural roughness (PC4), and background loudness (PC5).

In parallel, perceptual evaluations based on the ISO 12913 soundscape framework were conducted to assess ISOP and ISOE across different drive conditions. For most EVs, ISOP and ISOE scores varied by drive condition, with a tendency for ISOP to be lower and ISOE to be higher under the stationary condition (D0). When comparing forward pass-by at 20 km/h (D20) and reverse driving at 6 km/h (R6), D20 was generally perceived as more pleasant and more eventful. However, 17 EVs exhibited similar spherical distribution, making it difficult to clearly distinguish differences among vehicles.

A correlation analysis was conducted between the representative indicators of psychoacoustic parameters derived by PCA and ISOP/ISOE. The results revealed a significant association between loudness, fluctuation strength, and sharpness, and perceptual scores under certain drive conditions. Only F_{10} and N_{90} , which served as representative indicators of PC4 and PC5, were found to be significantly correlated with mean ISOP. However, not all correlations were significant across the entire set of AVAS sounds. However, nonlinear tendencies were observed in soundscape evaluation as psychoacoustic parameters changed. These results imply that the perceptual evaluation of AVAS sounds may involve more complex nonlinear structures.

HCA, based on the soundscape attribute scores and their condition-specific variations, revealed three distinct AVAS sound types. Cluster 1, characterized by sound features distinct from those of ICE vehicles, was associated with higher pleasantness scores, whereas Cluster 3, comprising EVs with repetitive warning tones during reverse driving, was associated with higher eventfulness scores. These findings demonstrate that differences in soundscape attribute scores can be linked to specific AVAS design approaches, highlighting the potential of AVAS to serve not only as a functional alert

system but also as a contributor to the auditory quality of urban environments. However, the observed inter-cluster differences also underscore the need for context-aware and type-specific AVAS design strategies.

This study represents one of the few attempts to evaluate AVAS using a soundscape-based framework. It provides an indirect yet meaningful perspective on the ways AVAS may influence urban auditory experiences. Nevertheless, the results also suggest that existing models and psychoacoustic parameters alone may not be sufficient to fully explain the perceptual qualities of AVAS sounds. Future research should aim to develop evaluation frameworks tailored to AVAS sounds and to expand the scope of analysis to include a wider range of vehicles, sound designs, and urban contexts.

ACKNOWLEDGMENTS

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts of interest to disclose.

Ethics Approval

This study was approved by the Institutional Review Board (IRB) of Chungnam National University (Approval No. 202402-SB-021-01). Informed consent was obtained from all participants.

DATA AVAILABILITY

556 The data that support the findings of this study are not publicly available due to restrictions but
557 may be available from the corresponding author upon reasonable request, subject to permission
558 from the data owner.
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No.		Drive condition	Mean Difference	SE	df_1	t	p	df_2	F	p	η_p^2
EV1	ISOP ^a	D0 – D20	-0.22	0.04	108	-5.16	< 0.001				
		D0 – R6	-0.26	0.05	108	-5.64	< 0.001	1.78	23.5	< 0.001	0.18
		D20 – R6	-0.04	0.03	108	-1.09	0.83				
	ISOE ^a	D0 – D20	0.08	0.03	102	2.58	0.03				
		D0 – R6	0.39	0.04	102	9.39	< 0.001	1.73	69.6	< 0.001	0.45
		D20 – R6	0.31	0.03	102	9.83	< 0.001				
	ISOP ^a	D0 – D20	-0.11	0.04	131	-2.58	0.03				
		D0 – R6	0.01	0.03	131	0.26	1.00	1.86	6.0	0.00	0.04
		D20 – R6	0.12	0.04	131	3.06	0.01				
EV2	ISOE	D0 – D20	0.00	0.04	130	0.10	1.00				
		D0 – R6	0.18	0.04	130	4.39	< 0.001	2	12.9	< 0.001	0.09
		D20 – R6	0.18	0.04	130	4.35	< 0.001				
	ISOP	D0 – D20	0.05	0.04	127	1.28	0.61				
		D0 – R6	0.13	0.04	127	3.32	0.00	2	5.5	0.00	0.04
		D20 – R6	0.08	0.04	127	1.98	0.15				
EV3	ISOE	D0 – D20	-0.19	0.04	128	-4.78	< 0.001				
		D0 – R6	0.08	0.05	128	1.77	0.24	2	22.4	< 0.001	0.15
		D20 – R6	0.27	0.04	128	6.92	< 0.001				
	ISOP	D0 – D20	-0.19	0.05	106	-3.87	< 0.001				
		D0 – R6	-0.13	0.05	106	-2.67	0.03	2	8.2	< 0.001	0.07
		D20 – R6	0.06	0.05	106	1.30	0.59				

No.		Drive condition	Mean Difference	<i>SE</i>	<i>df</i> ₁	<i>t</i>	<i>p</i>	<i>df</i> ₂	<i>F</i>	<i>p</i>	η_p^2
	ISOE ^a	D0 – D20	0.07	0.02	86	3.06	0.01	1.73	12.4	< 0.001	0.13
		D0 – R6	0.16	0.03	86	4.63	< 0.001				
		D20 – R6	0.08	0.04	86	2.36	0.06				
EV5	ISOP	D0 – D20	-0.03	0.04	116	-0.76	1.00	2	3.0	0.05	0.03
		D0 – R6	0.07	0.04	116	1.56	0.37				
		D20 – R6	0.10	0.04	116	2.52	0.04				
	ISOE ^a	D0 – D20	-0.16	0.03	111	-4.96	< 0.001	1.85	19.3	< 0.001	0.15
		D0 – R6	0.06	0.04	111	1.51	0.40				
		D20 – R6	0.22	0.04	111	6.16	< 0.001				
	ISOP	D0 – D20	0.00	0.04	105	0.09	1.00	2	0.86	0.42	0.01
		D0 – R6	-0.04	0.04	105	-1.08	0.85				
		D20 – R6	-0.04	0.03	105	-1.32	0.57				
	ISOE	D0 – D20	-0.02	0.03	97	-0.54	1.00	2	38.9	< 0.001	0.29
		D0 – R6	0.26	0.04	97	6.87	< 0.001				
		D20 – R6	0.28	0.04	97	7.69	< 0.001				
EV7	ISOP	D0 – D20	-0.18	0.04	124	-4.11	< 0.001	2	30.3	< 0.001	0.20
		D0 – R6	-0.33	0.04	124	-7.57	< 0.001				
		D20 – R6	-0.15	0.04	124	-3.76	< 0.001				
	ISOE ^a	D0 – D20	0.12	0.03	121	3.98	< 0.001	1.75	58.4	< 0.001	0.33
		D0 – R6	0.40	0.04	121	9.27	< 0.001				
		D20 – R6	0.28	0.04	121	7.08	< 0.001				
EV8	ISOP	D0 – D20	-0.32	0.05	131	-6.64	< 0.001	2	24.9	< 0.001	0.16
		D0 – R6	-0.09	0.05	131	-1.91	0.18				
		D20 – R6	0.23	0.05	131	4.96	< 0.001				
	ISOE ^a	D0 – D20	0.04	0.04	131	1.10	0.83	1.93	1.53	0.22	0.01

No.	Drive condition	Mean Difference	SE	df_1	t	p	df_2	F	p	η_p^2
EV9	ISOP	D0 – R6	0.07	131	1.58	0.35				
		D20 – R6	0.03	131	0.76	1.00				
		D0 – D20	0.02	122	0.32	1.00				
	ISOE	D0 – R6	-0.08	122	-2.40	0.05	1.76	3.56	0.03	0.03
		D20 – R6	-0.10	122	-2.70	0.02				
		D0 – D20	0.04	119	1.03	0.92				
EV10	ISOP	D20 – R6	-0.01	122	-0.20	0.84	1	0.04	0.84	0.00
	ISOE	D20 – R6	0.37	118	9.21	< 0.001	1	84.7	< 0.001	0.42
EV11	ISOP	D20 – R6	-0.00	129	-0.21	0.84	1	0.04	0.84	0.00
	ISOE	D20 – R6	0.29	129	6.41	< 0.001	1	41.1	< 0.001	0.24
EV12	ISOP	D20 – R6	-0.14	121	-3.05	0.00	1	9.33	0.00	0.07
	ISOE	D20 – R6	-0.11	115	-3.05	0.00	1	9.33	0.00	0.08
EV13	ISOP	D20 – R6	0.13	123	2.99	0.00	1	8.96	0.00	0.07
	ISOE	D20 – R6	0.09	124	2.57	0.01	1	6.6	0.01	0.05
EV14	ISOP	D20 – R6	0.21	131	4.56	< 0.001	1	20.8	< 0.001	0.14
	ISOE	D20 – R6	-0.03	131	-0.79	0.43	1	0.62	0.43	0.01
EV15	ISOP	D20 – R6	0.06	130	1.25	0.21	1	1.57	0.21	0.01
	ISOE	D20 – R6	0.10	130	2.46	0.02	1	6.1	0.02	0.05
EV16	ISOP	D20 – R6	0.10	131	2.20	0.03	1	4.84	0.03	0.04
	ISOE	D20 – R6	0.06	127	1.79	0.08	1	3.2	0.08	0.03
EV17	ISOP	D20 – R6	0.17	118	3.33	0.01	1	11.1	0.00	0.09
	ISOE	D20 – R6	-0.03	114	-0.98	0.33	1	0.97	0.33	0.01

^a Assumption of sphericity was violated, and Huynh-Feldt correction was applied.

REFERENCES

- ¹M. S. Alam, and R. M. Martinez, “Pedestrian crossing behaviour in front of electric vehicles emitting synthetic sounds,” in *Proceedings of 54th International Congress & Exposition on Noise Control Engineering. São Paulo, Brazil* (2025) p. 12. doi:10.4121/629CAE37-76E7-4B14-8693-25C96A263B4B
- ²F. Aletta, B. De Coensel, and P. M. Lindborg, “Editorial: Human perception of environmental sounds,” *Front. Psychol.* **12**, 714591 (2021). doi:10.3389/FPSYG.2021.714591/BIBTEX
- ³F. Aletta, J. Kang, and Ö. Axelsson, “Soundscape descriptors and a conceptual framework for developing predictive soundscape models,” *Landsc. Urban Plan.* **149**, 65–74 (2016). doi:10.1016/J.LANDURBPLAN.2016.02.001
- ⁴J. P. Arenas, “On the impact of electric vehicle transition on urban noise pollution,” *Curr. Opin. Environ. Sci. Heal.* **45**, 100623 (2025). doi:10.1016/J.COESH.2025.100623
- ⁵Z. Azkorra, G. Pérez, J. Coma, L. F. Cabeza, S. Bures, J. E. Álvaro, A. Erkoreka, et al., “Evaluation of green walls as a passive acoustic insulation system for buildings,” *Appl. Acoust.* **89**, 46–56 (2015). doi:10.1016/J.APACOUST.2014.09.010
- ⁶P. Bazilinskyy, R. Merino-Martínez, E. Özcan, D. Dodou, and J. de Winter, “Exterior sounds for electric and automated vehicles: Loud is effective,” *Appl. Acoust.* **214**, 109673 (2023). doi:10.1016/J.APACOUST.2023.109673
- ⁷M. J. Blanca, J. Arnau, F. J. García-Castro, R. Alarcón, and R. Bono, “Repeated measures ANOVA and adjusted F-tests when sphericity is violated: which procedure is best?,” *Front. Psychol.* **14**, 1192453 (2023). doi:10.3389/FPSYG.2023.1192453/BIBTEX
- ⁸H. Campello-Vicente, R. Peral-Orts, N. Campillo-Davo, and E. Velasco-Sanchez, “The effect of electric vehicles on urban noise maps,” *Appl. Acoust.* **116**, 59–64 (2017). doi:10.1016/J.APACOUST.2016.09.018

587 ⁹P. Cocron, and J. F. Krems, “Driver perceptions of the safety implications of quiet electric
 588 vehicles,” *Accid. Anal. Prev.* **58**, 122–131 (2013). doi:10.1016/J.AAP.2013.04.028
 589 ¹⁰B. De Coensel, S. Vanwetswinkel, and D. Botteldooren, “Effects of natural sounds on the
 590 perception of road traffic noise,” *J. Acoust. Soc. Am.* **129**, EL148–EL153 (2011).
 591 doi:10.1121/1.3567073
 592 ¹¹Deutsches Institut für Normung, *DIN 45692: Measurement technique for the simulation of the auditory*
 593 *sensation of sharpness*, (Berlin, Germany, 2009).
 594 ¹²Ecma International, “ECMA 418-2 — Psychoacoustic metrics for ITT equipment — Part 2
 595 (models based on human perception) (1st ed.),”
 596 ¹³M. S. Engel, A. Fiebig, C. Pfaffenbach, and J. Fels, “A review of the use of psychoacoustic
 597 indicators on soundscape studies,” *Curr. Pollut. Reports* **7**, 359–378 (2021).
 598 doi:10.1007/S40726-021-00197-1/TABLES/2
 599 ¹⁴J. Fagerlön, A. Sirkka, R. Johnsson, and S. Lindberg, “Acoustic vehicle alerting systems: Will they
 600 affect the acceptance of electric vehicles?,” doi: 10.1145/3243274.3243305.
 601 doi:10.1145/3243274.3243305
 602 ¹⁵H. Fastl, and E. Zwicker, *Psychoacoustics: facts and models*, (Springer Science & Business Media, 2006),
 603 Vol. 22.
 604 ¹⁶A. Fiebig, “Electric vehicles get alert signals to be heard by pedestrians: benefits and drawbacks,”
 605 *Acoust. Today* **16**, 20–28 (2020). Retrieved from
 606 [https://www.semanticscholar.org/paper/Electric-Vehicles-Get-Alert-Signals-to-be-Heard-by-](https://www.semanticscholar.org/paper/Electric-Vehicles-Get-Alert-Signals-to-be-Heard-by-Fiebig/f2d53d4a1a07f91c10d7e063f5ea503d645445f8)
 607 [Fiebig/f2d53d4a1a07f91c10d7e063f5ea503d645445f8](https://www.semanticscholar.org/paper/Electric-Vehicles-Get-Alert-Signals-to-be-Heard-by-Fiebig/f2d53d4a1a07f91c10d7e063f5ea503d645445f8)
 608 ¹⁷S. Fleury, É. Jamet, V. Roussarie, L. Bosc, and J. C. Chamard, “Effect of additional warning sounds
 609 on pedestrians’ detection of electric vehicles: An ecological approach,” *Accid. Anal. Prev.* **97**,
 610 176–185 (2016). doi:10.1016/J.AAP.2016.09.002

611 ¹⁸A. Fortino, L. Eckstein, J. Viehöfer, and J. Pampel, “Acoustic Vehicle Alerting Systems (AVAS) -
612 Regulations, realization and sound design challenges,” SAE Int. J. Passeng. Cars - Mech. Syst.
613 **9**, 995–1003 (2016). doi:10.4271/2016-01-1784

614 ¹⁹Z. Han, J. Kang, and Q. Meng, “A time series analysis model of the relationship between
615 psychoacoustic parameters of urban soundscape spatial sequences and emotional changes,” J.
616 Acoust. Soc. Am. **152**, 2022–2037 (2022). doi:10.1121/10.0014287

617 ²⁰M. Hirz, and T. T. Nguyen, “Life-cycle CO2-equivalent emissions of cars driven by conventional
618 and electric propulsion systems,” World Electr. Veh. J. 2022, Vol. 13, Page 61 **13**, 61 (2022).
619 doi:10.3390/WEVJ13040061

620 ²¹J. Y. Hong, and J. Y. Jeon, “Comparing associations among sound sources, human behaviors, and
621 soundscapes on central business and commercial streets in Seoul, Korea,” , doi:
622 10.1016/j.buildenv.2020.107327. doi:10.1016/j.buildenv.2020.107327

623 ²²M. C. Hsieh, H. J. Chen, M. Le Tong, and C. W. Yan, “Effect of Environmental Noise, Distance
624 and Warning Sound on Pedestrians’ Auditory Detectability of Electric Vehicles,” Int. J.
625 Environ. Res. Public Health **18**, 9290 (2021). doi:10.3390/IJERPH18179290

626 ²³ISO 12913-1, *Acoustics—Soundscape—Part 1: Definition and conceptual framework*, (Geneva, Switzerland,
627 2014).

628 ²⁴ISO 12913-2, *Acoustics - Soundscape - Part 2: Data Collection and Reporting Requirements*, (Geneva,
629 Switzerland, 2018), Vol. 44.

630 ²⁵ISO 12913-3, *Acoustics — Soundscape — Part 3 Data analysis*, (Geneva, Switzerland, 2019).

631 ²⁶ISO 16254, *Acoustics — Measurement of sound emitted by road vehicles of category M and N at standstill and*
632 *low speed operation — Engineering method*, (Geneva, Switzerland, 2024).

633 ²⁷H. I. Jo, and J. Y. Jeon, “The influence of human behavioral characteristics on soundscape
634 perception in urban parks: Subjective and observational approaches,” *Landsc. Urban Plan.* **203**,
635 103890 (2020). doi:10.1016/J.LANDURBPLAN.2020.103890

636 ²⁸H. I. Jo, and J. Y. Jeon, “Urban soundscape categorization based on individual recognition,
637 perception, and assessment of sound environments,” *Landsc. Urban Plan.* **216**, 104241 (2021).
638 doi:10.1016/J.LANDURBPLAN.2021.104241

639 ²⁹T. H. Kim, G. H. Kim, and J. Y. Hong, “Translating perceived affective quality attributes of
640 soundscape from English into Korean,” *Appl. Acoust.* **222**, 110051 (2024).
641 doi:10.1016/J.APACOUST.2024.110051

642 ³⁰F. Laib, and J. Alexander Schmidt, “Acoustic Vehicle Alerting Systems (AVAS) of electric cars and
643 its possible influence on urban soundscape,” *Proc. Int. Congr. Acoust.* **2019-Sept**, 2544–2551
644 (2019). doi:10.18154/RWTH-CONV-239469

645 ³¹L. Maffei, and M. Masullo, “Electric vehicles and urban noise control policies,” *Arch. Acoust.* **39**,
646 333–341 (2015). doi:10.2478/AOA-2014-0038

647 ³²A. Mitchell, F. Aletta, ; Tin, ; O., J. Kang, and T. Oberman, “Soundscape perception indices (SPIs):
648 Developing context-dependent single value scores of multidimensional soundscape perceptual
649 quality,” *J. Acoust. Soc. Am.* **156**, 3694–3706 (2024). doi:10.1121/10.0034417

650 ³³L. Müller, J. Forssén, W. Kropp, L. M€, J. Forss, and F. Aletta, “Auditory localization of multiple
651 stationary electric vehicles,” *J. Acoust. Soc. Am.* **157**, 2029–2041 (2025).
652 doi:10.1121/10.0036248

653 ³⁴L. Müller, and W. Kropp, “Auralization of electric vehicles for the perceptual evaluation of
654 acoustic vehicle alerting systems,” *Acta Acust.* **8**, 27 (2024). doi:10.1051/aacus/2024025

655 ³⁵M. Munder, R. Vitale, and C. C. Carbon, “Back to the future: Innovativeness in sound design for
656 electrified vehicles,” , doi: 10.1177/20416695251323786. doi:10.1177/20416695251323786

³⁶Y. Nakajima, and H. Shimamoto, “Mathematical model to evaluate the impact of power supply
 constraint for electric vehicles on transportation network,” *Int. J. Intell. Transp. Syst. Res.* **23**,
 16–28 (2025). doi:10.1007/S13177-024-00429-W/TABLES/9

³⁷T. Nguyen, and M. Morinaga, “Effect of roadside trees on pedestrians’ psychological evaluation of
 traffic noise,” , doi: 10.3389/FPSYG.2023.1166318. doi:10.3389/FPSYG.2023.1166318

³⁸T. Oberman, A. Latini, F. Aletta, G. Gozzi, J. Kang, and S. Torresin, “Human sounds and
 associated tonality disrupting perceived soundscapes in protected natural areas,” , doi:
 10.1038/S41598-025-08524-Y. doi:10.1038/S41598-025-08524-Y

³⁹M. A. Pallas, M. Capatina, A. Le Bellec, S. Bianchetti, and J. Cesbron, “Environmental noise from
 alerting signals (AVAS) of electric vehicles: Comparison with noise prediction models at urban
 speeds,” , doi: 10.61782/fa.2023.0613. doi:10.61782/fa.2023.0613

⁴⁰E. Parizet, and R. Robart, “Detectability and annoyance of warning sounds for electric vehicles,” *J.*
Acoust. Soc. Am. **133**, 3323–3323 (2013). doi:10.1121/1.4805556

⁴¹E. Parizet, R. Robart, P. Pondrom, J. C. Chamard, G. Baudet, D. Quinn, K. Janssens, et al.,
 “Additional efficient warning sounds for electric and hybrid vehicles,” , doi:
 10.1002/9781119307761.CH32. doi:10.1002/9781119307761.CH32

⁴²J.-F. Petiot, K. Legeay, and M. Lagrange, “Optimization of the sound of electric vehicles according
 to unpleasantness and detectability,” in *Proceedings of the Design Society: International Conference on*
Engineering Design (Cambridge University Press, 2019) pp. 3949–3958.

⁴³F. G. Pratico, P. G. Briante, and G. Speranza, “Acoustic impact of electric vehicles,” in *20th IEEE*
Mediterranean Electrotechnical Conference (MELECON, IEEE, 2020) pp. 7–12.
 doi:10.1109/MELECON48756.2020.9140669

⁴⁴F. G. Praticò, and R. Fedele, “Electric vehicles diffusion: Changing pavement acoustic design?,”
Noise Mapp. **8**, 281–294 (2021). doi:10.1515/NOISE-2021-0023/PDF

681 ⁴⁵A. R. Rodrigues Ferraz Esteves, E. M. Campos Magalhães, and G. Bernardes De Almeida, “Sound
682 Design for Electric Vehicles: Enhancing Safety and User Experience Through Acoustic Vehicle
683 Alerting System (AVAS).,” doi:10.4271/2025-01-0068

684 ⁴⁶I. Salleh, M. Z. Md. Zain, and R. I. Raja Hamzah, “Evaluation of annoyance and suitability of a
685 back-up warning sound for electric vehicles,” *Int. J. Automot. Mech. Eng.* **8**, 1267–1277
686 (2013). doi:10.15282/IJAME.8.2013.16.0104

687 ⁴⁷A. Terenzi, S. Spinsante, and S. Cecchi, “Review on electric vehicles exterior noise generation and
688 evaluation,” , doi: 10.23919/aeitautomotive50086.2020.9307397.
689 doi:10.23919/aeitautomotive50086.2020.9307397

690 ⁴⁸T. Walton, D. S. Elias, and A. J. Torija, “Auditory alerts for e-scooters: Relationship between
691 pedestrian auditory detection rates and alert sound level,” *Appl. Acoust.* **214**, 109683 (2023).
692 doi:10.1016/J.APACOUST.2023.109683

693 ⁴⁹T. Walton, A. J. Torija, and A. S. Elliott, “Development of electric scooter alerting sounds using
694 psychoacoustical metrics,” *Appl. Acoust.* **201**, 109136 (2022).
695 doi:10.1016/J.APACOUST.2022.109136

696 ⁵⁰M. S. Wogalter, R. W. Lim, and P. G. Nyeste, “On the hazard of quiet vehicles to pedestrians and
697 drivers,” *Appl. Ergon.* **45**, 1306–1312 (2014). doi:10.1016/J.APERGO.2013.08.002

698 ⁵¹D. Yang, X. Cao, and Q. Meng, “Effects of a human sound-based index on the soundscapes of
699 urban open spaces,” *Sci. Total Environ.* **802**, 149869 (2022).
700 doi:10.1016/J.SCITOTENV.2021.149869

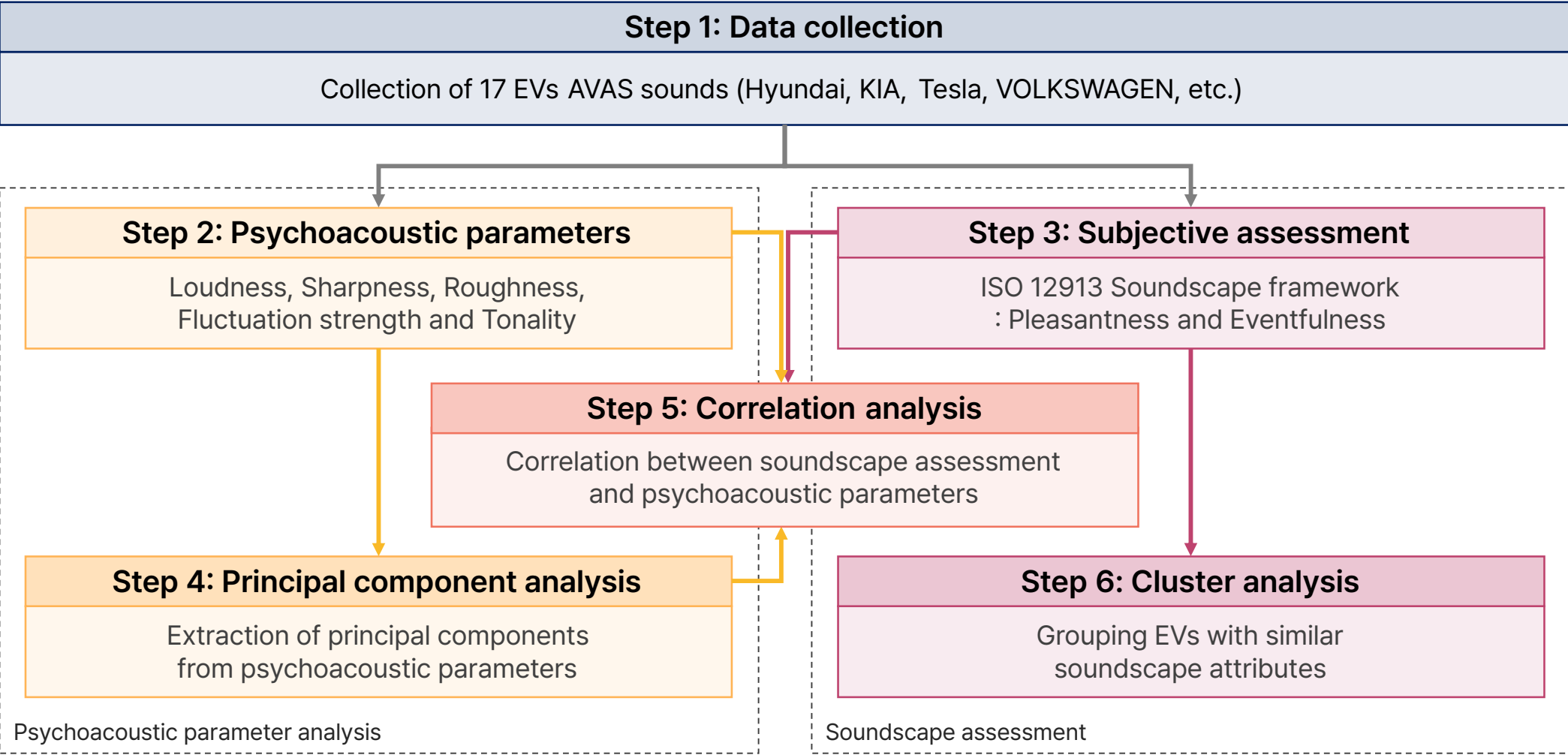
701 ⁵²X. Yang, G. Zhang, X. Lu, Y. Zhang, and J. Kang, “Contribution of soundscape appropriateness
702 to soundscape quality assessment in space: A mediating variable affecting acoustic comfort,” *J.*
703 *Environ. Manage.* **372**, 123321 (2024). doi:10.1016/J.JENVMAN.2024.123321

704 ⁵³J. You, and J. Y. Jeon, “Just noticeable differences in sound quality metrics for refrigerator noise,”
705 Noise Control Eng. J. **56**, 414 (2008). doi:10.3397/1.2987734

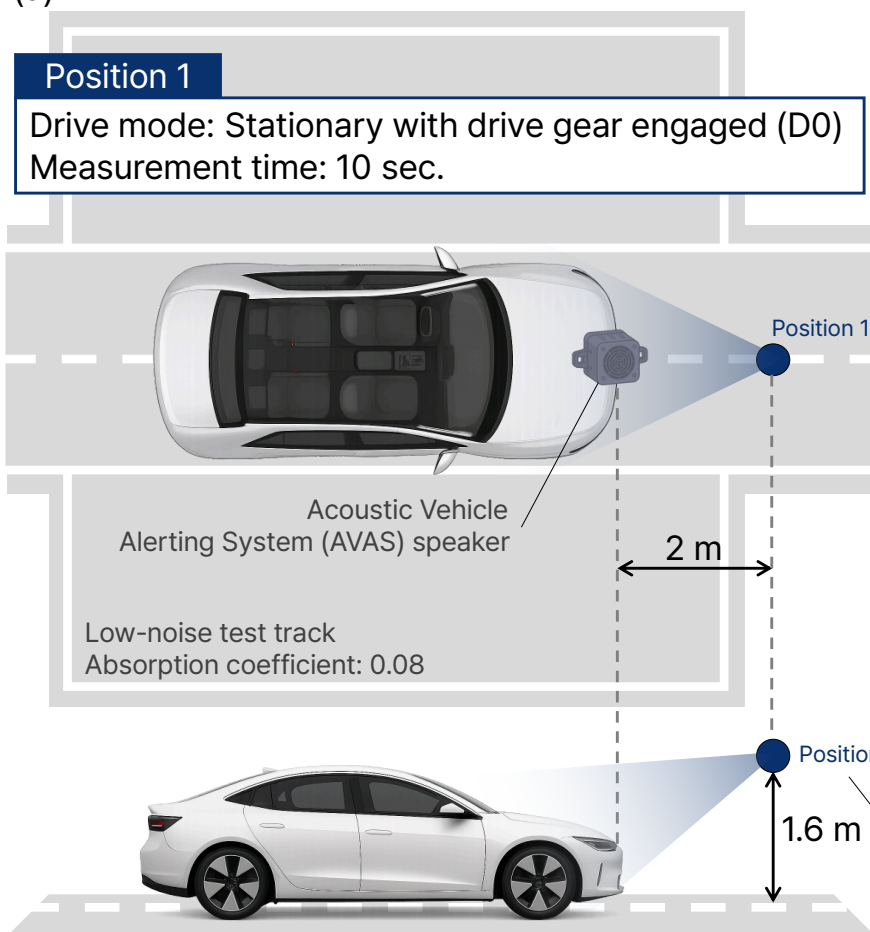
706 ⁵⁴R. Zaino, V. Ahmed, A. M. Alhammadi, and M. Alghoush, “Electric vehicle adoption: A
707 comprehensive systematic review of technological, environmental, organizational and policy
708 impacts,” World Electr. Veh. J. 2024, Vol. 15, Page 375 **15**, 375 (2024).
709 doi:10.3390/WEVJ15080375

710 ⁵⁵S. Zhang, and L. Chen, “Acoustic information masking effects of natural sounds on traffic noise
711 based on psychological health in open urban spaces,” Front. Public Heal. **11**, 1031501 (2023).
712 doi:10.3389/FPUBH.2023.1031501/BIBTEX

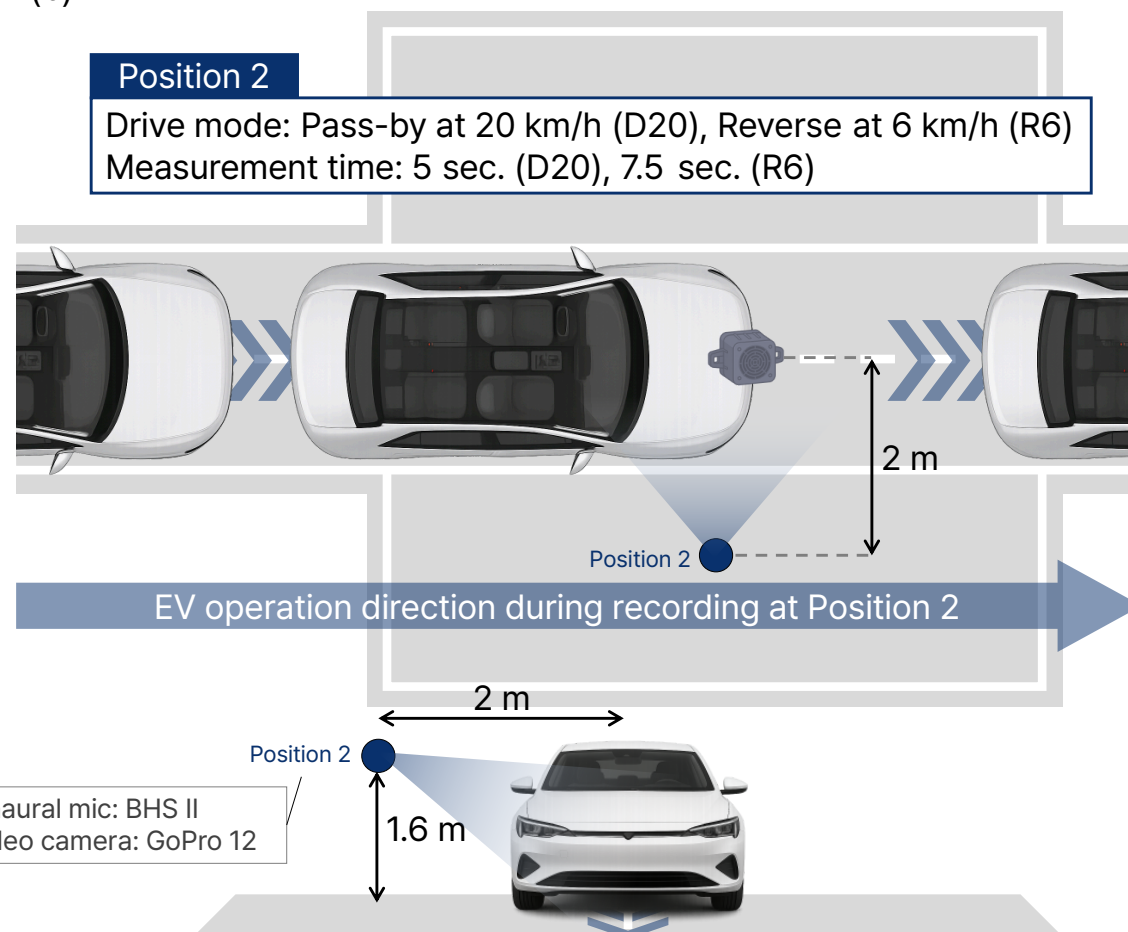
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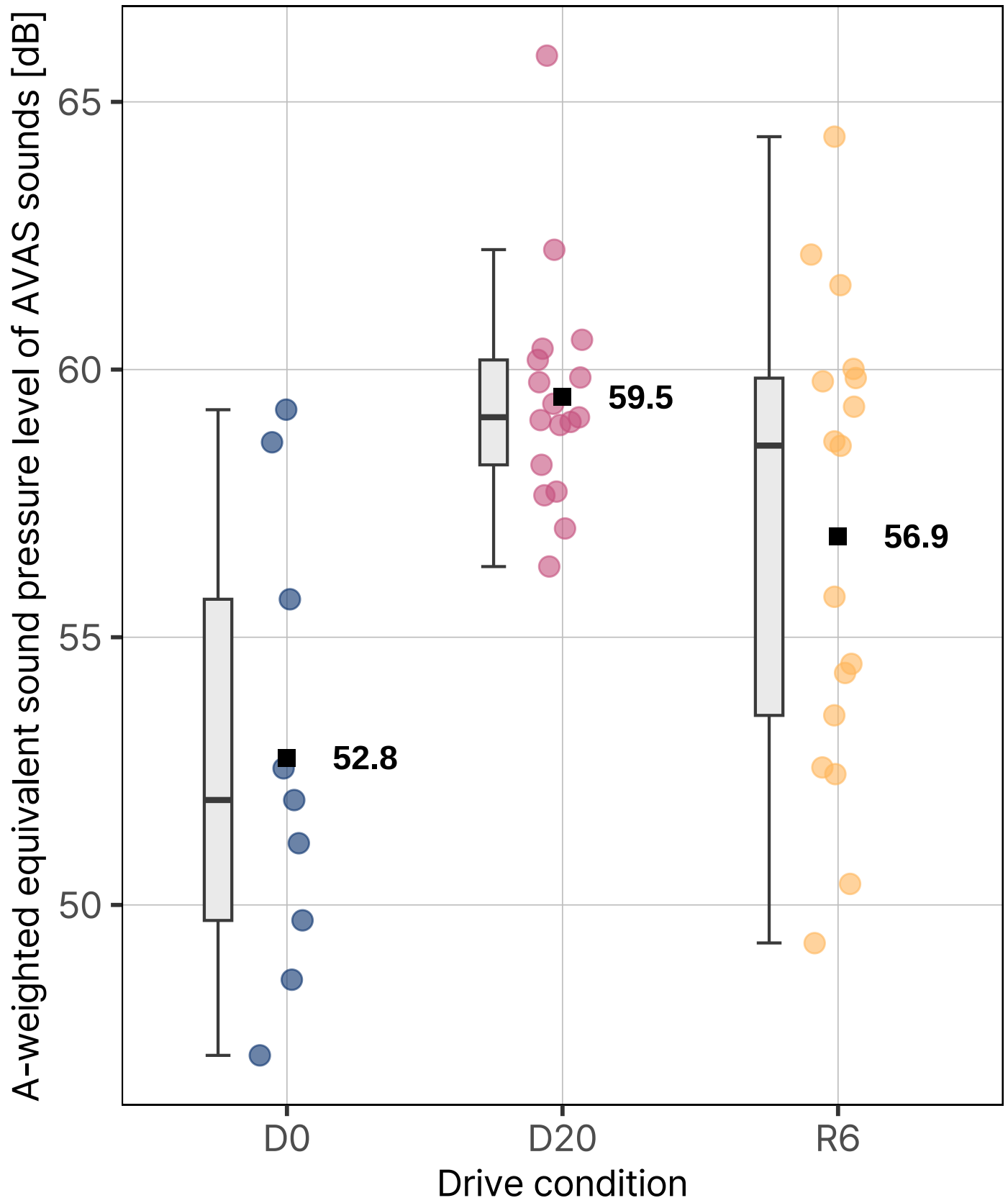


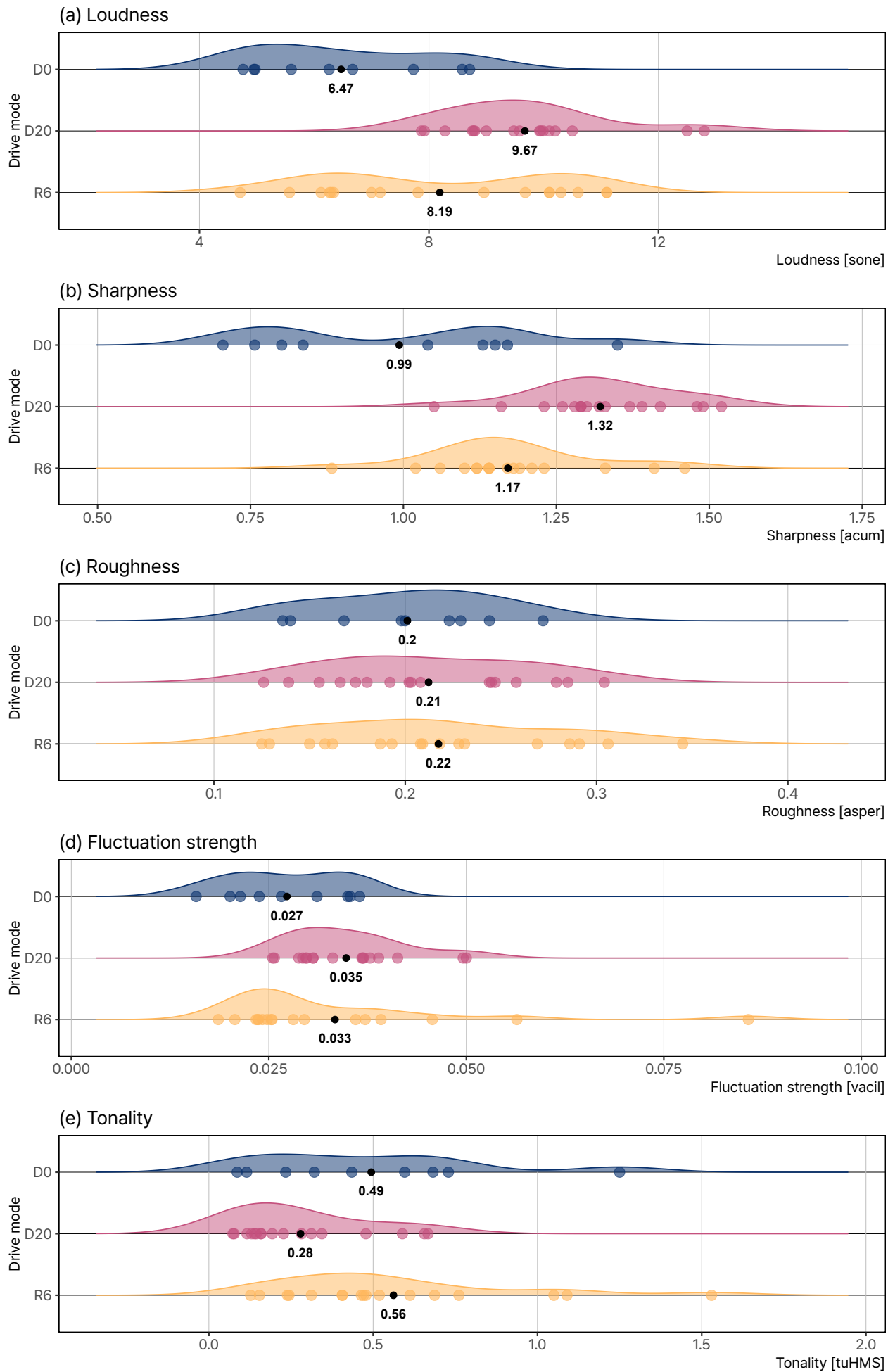
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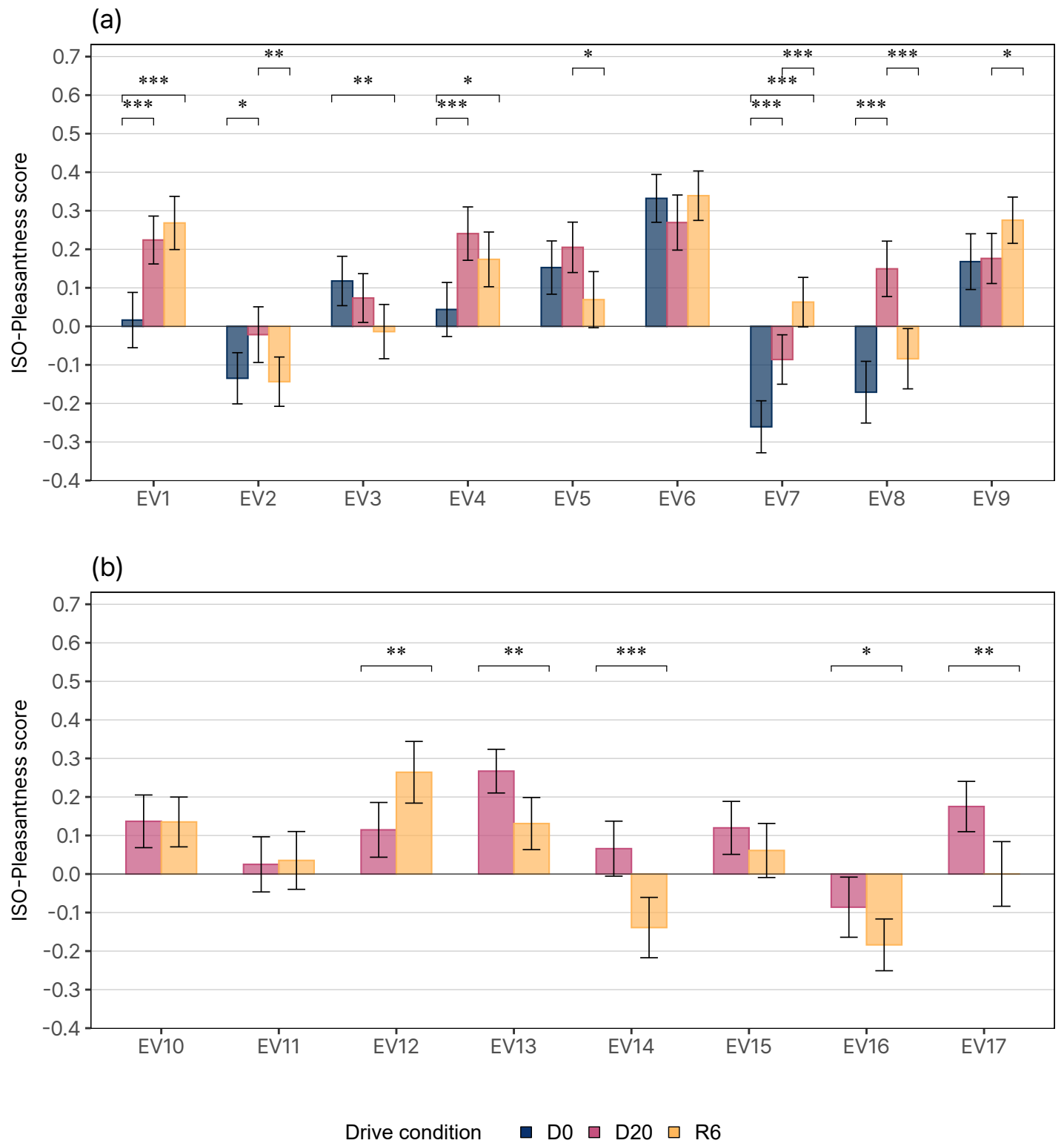


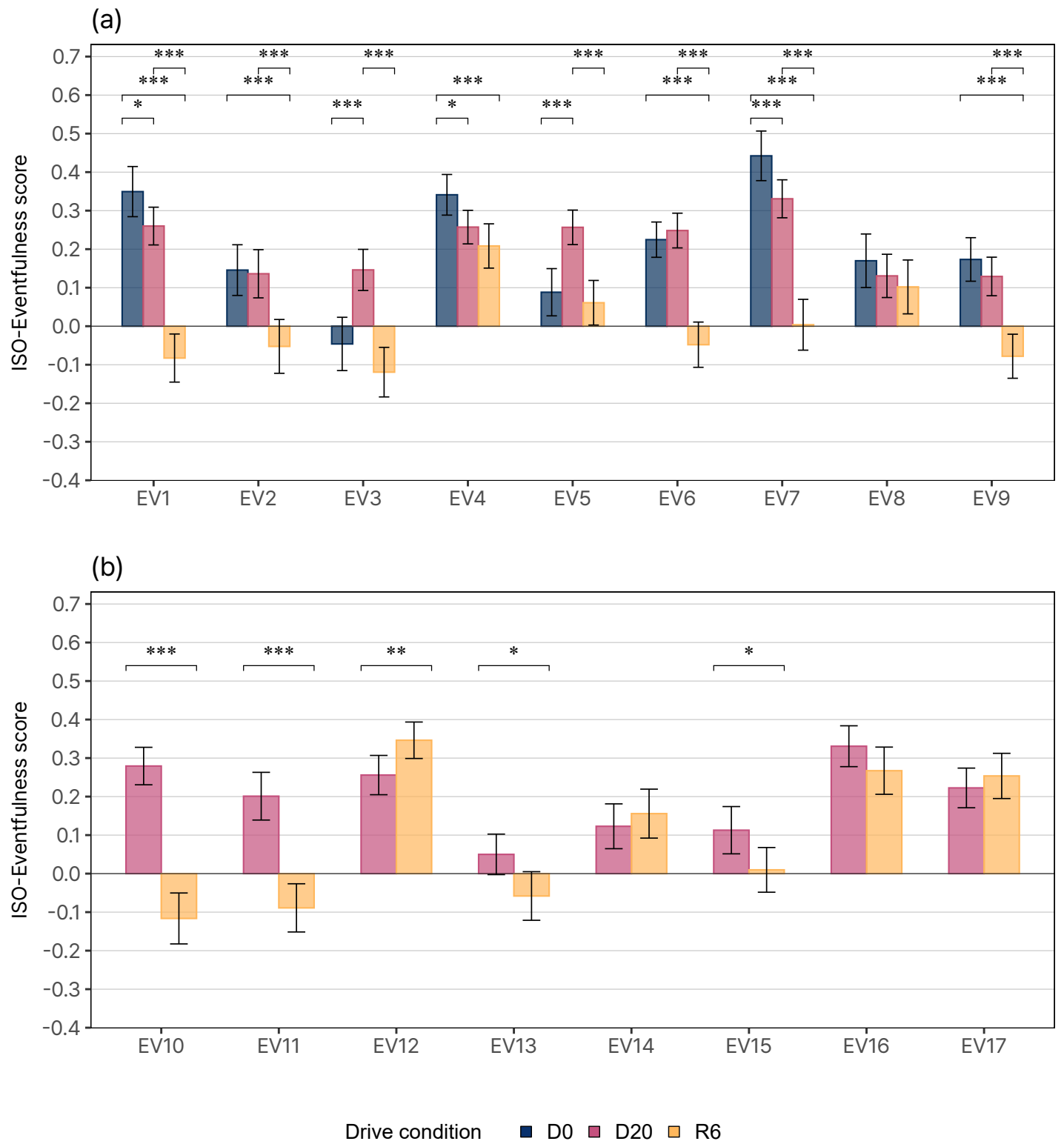
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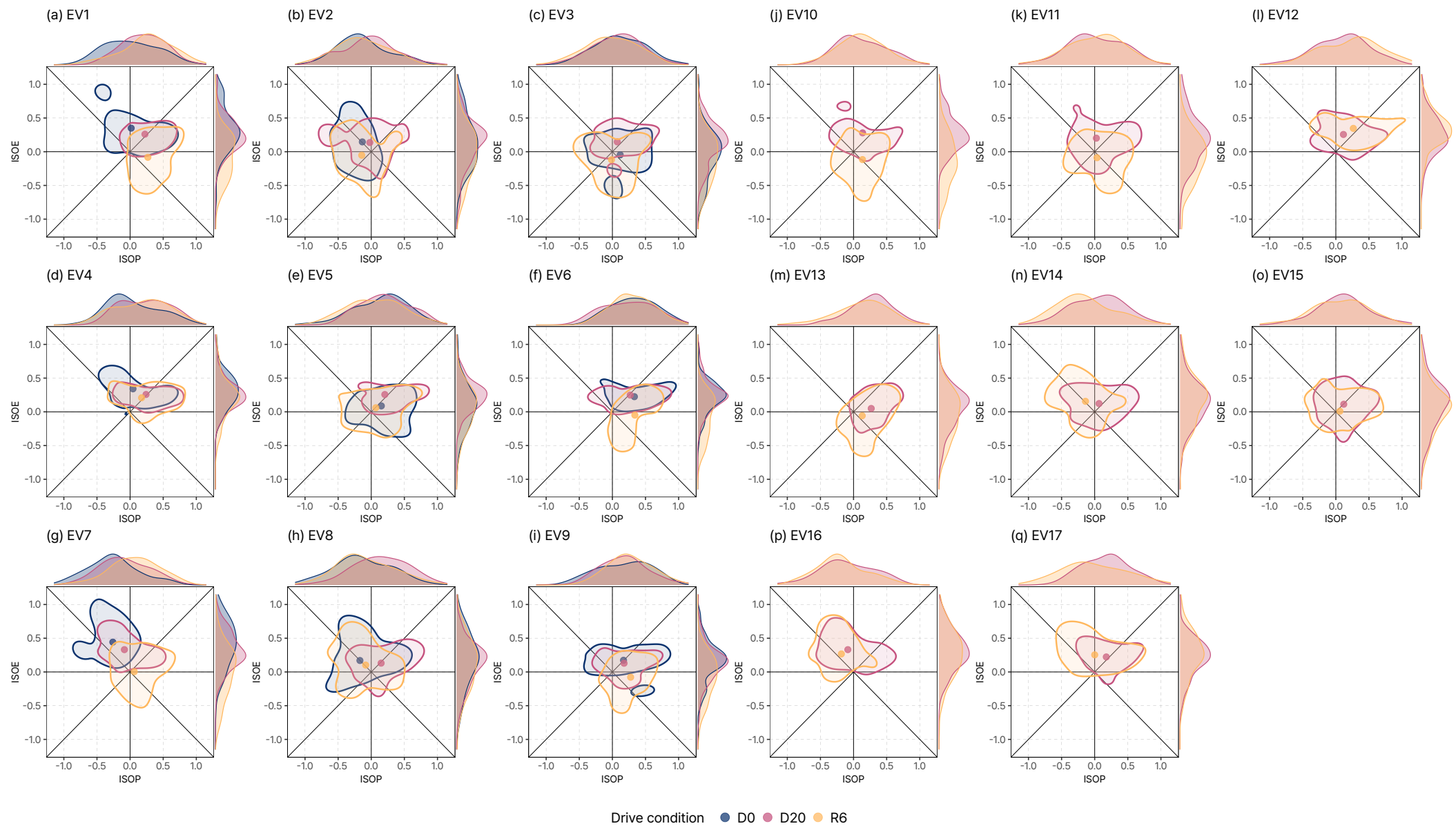


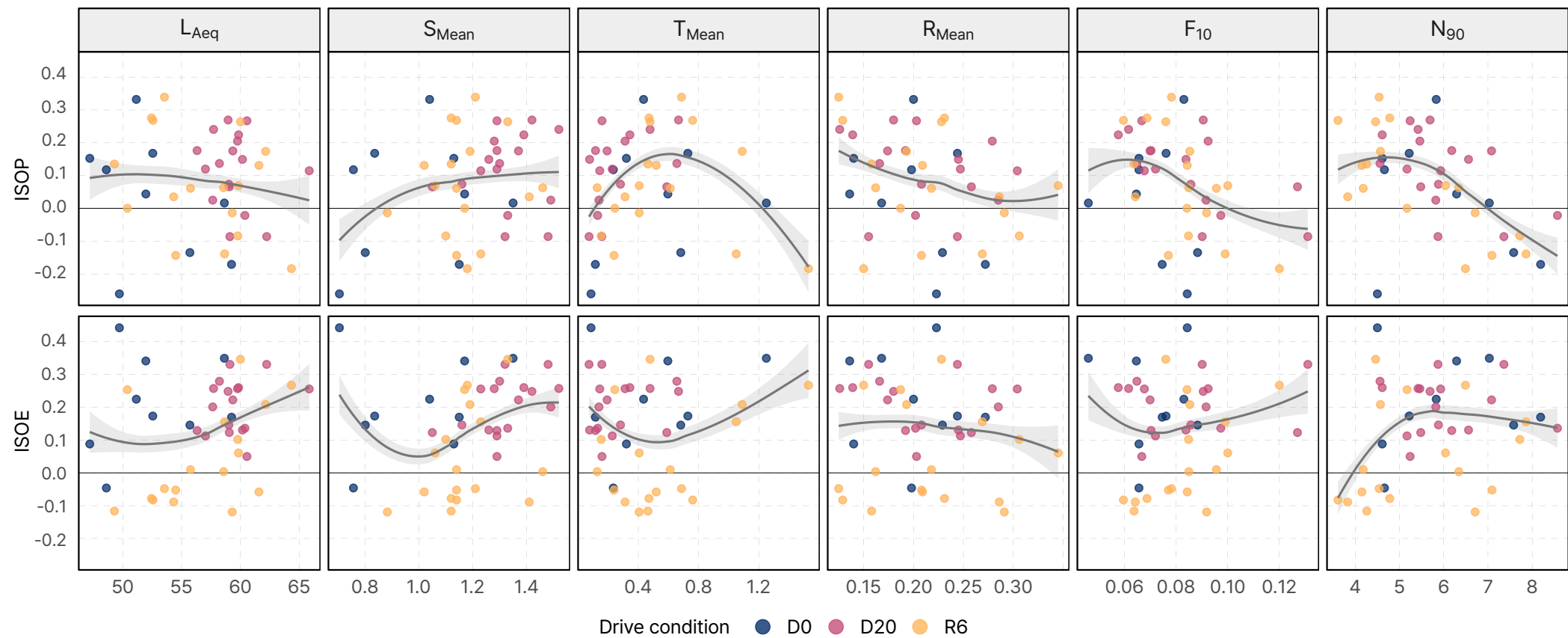


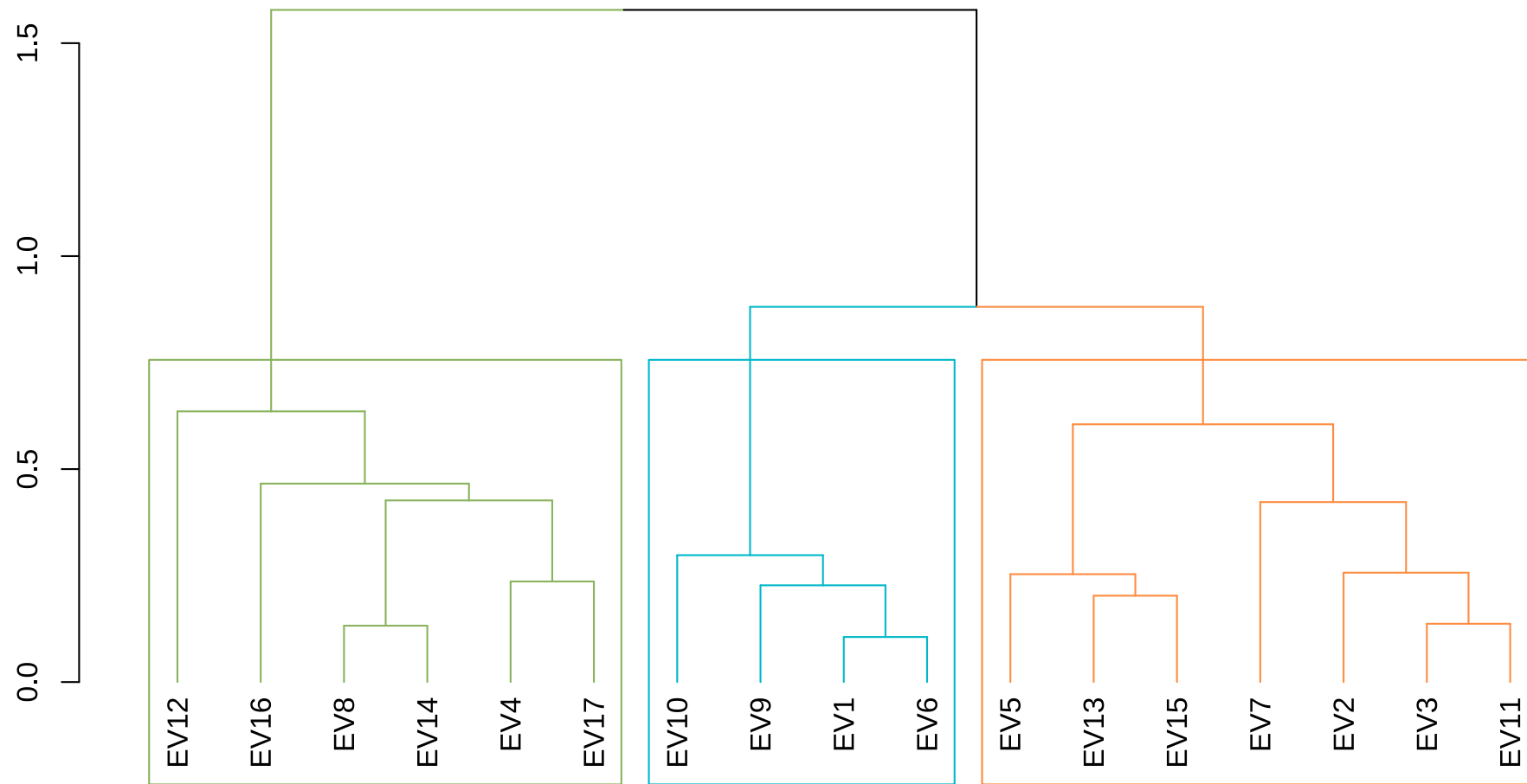


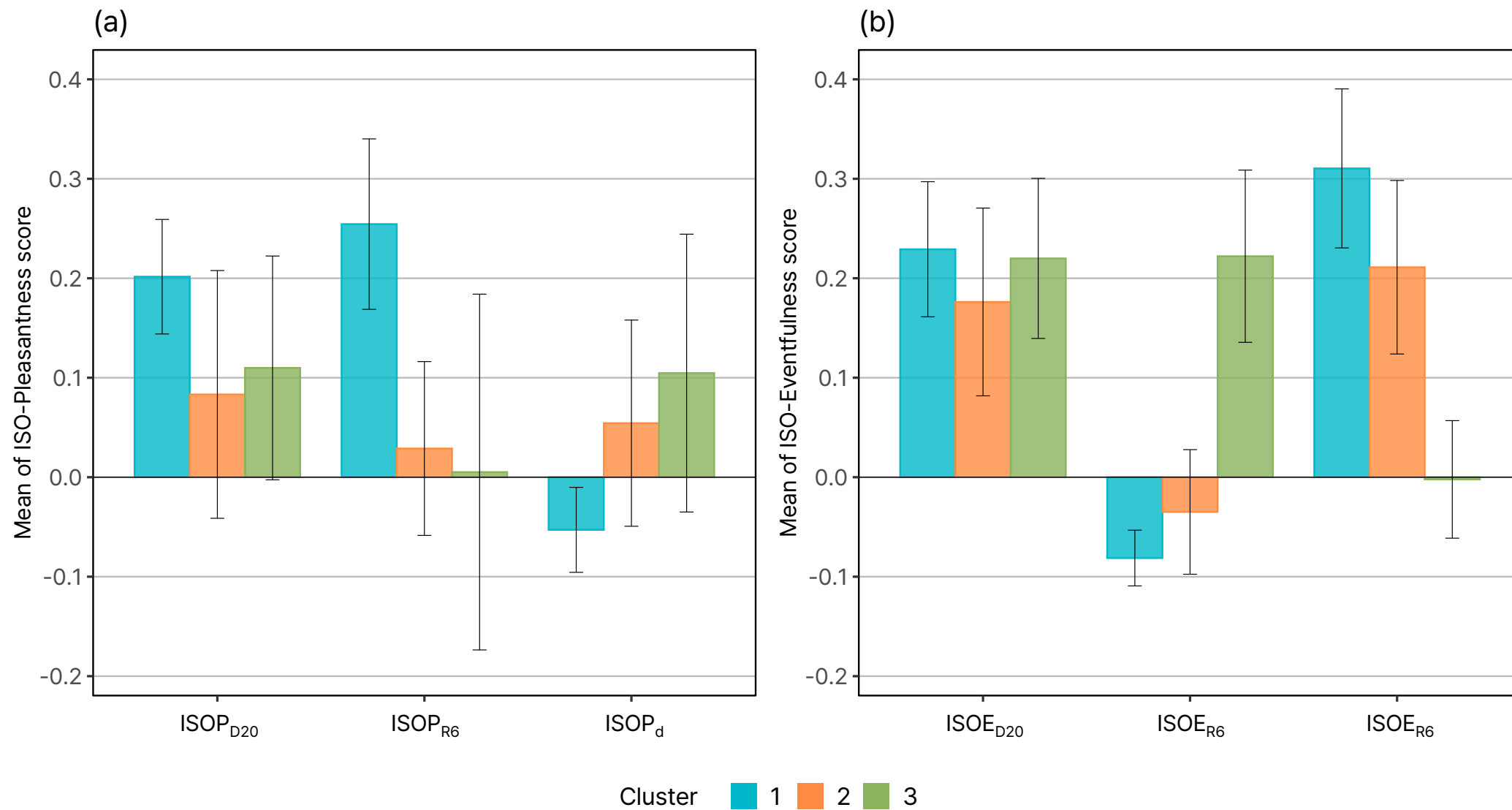


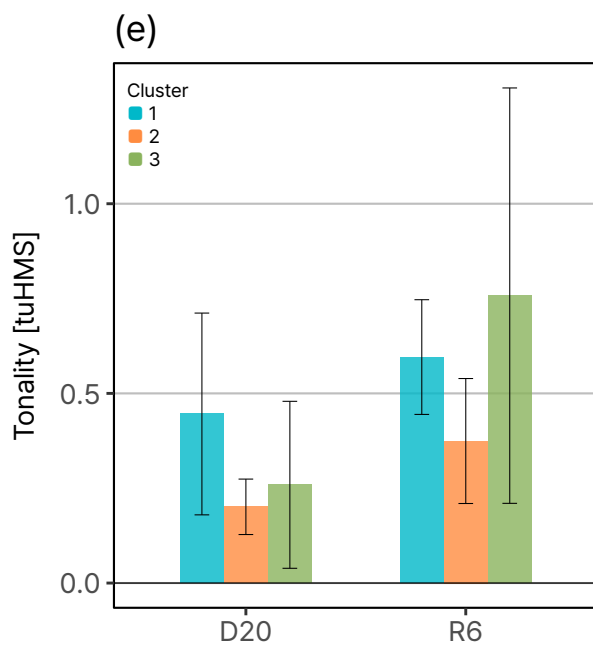
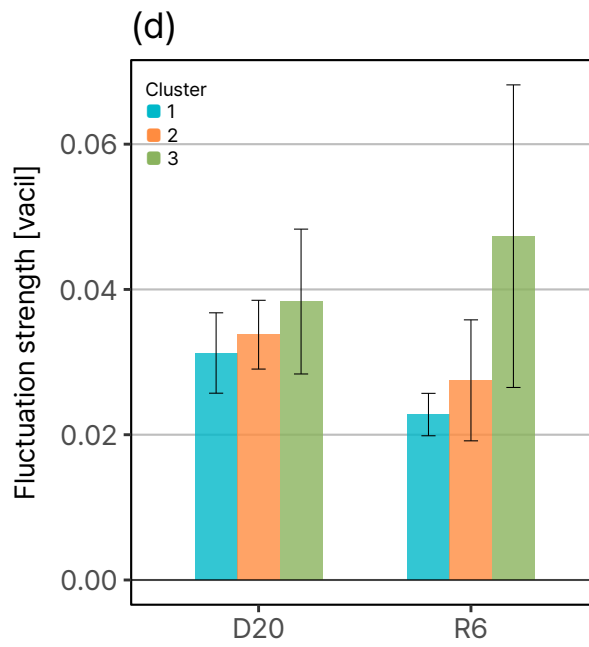
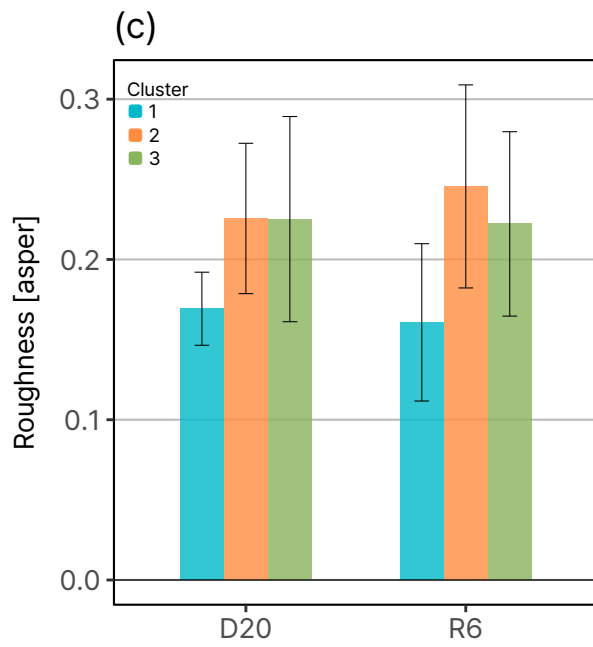
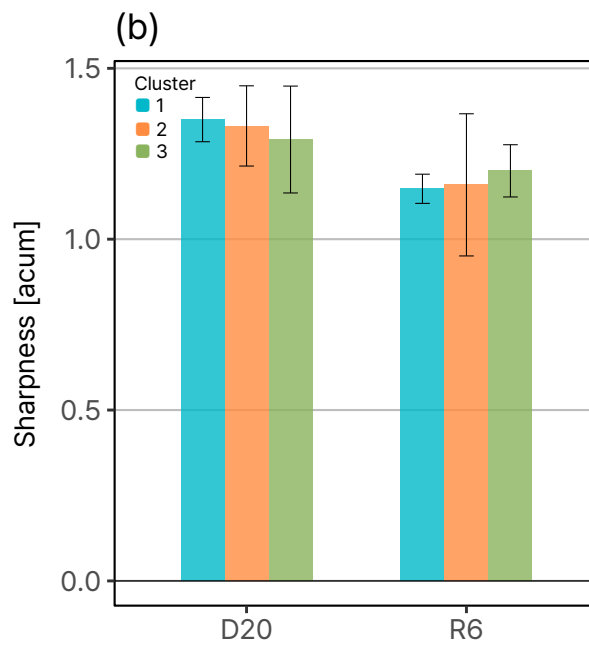
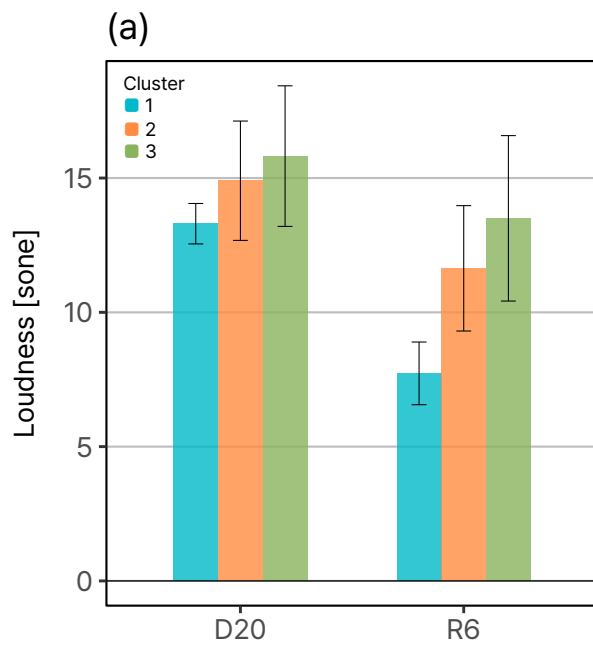


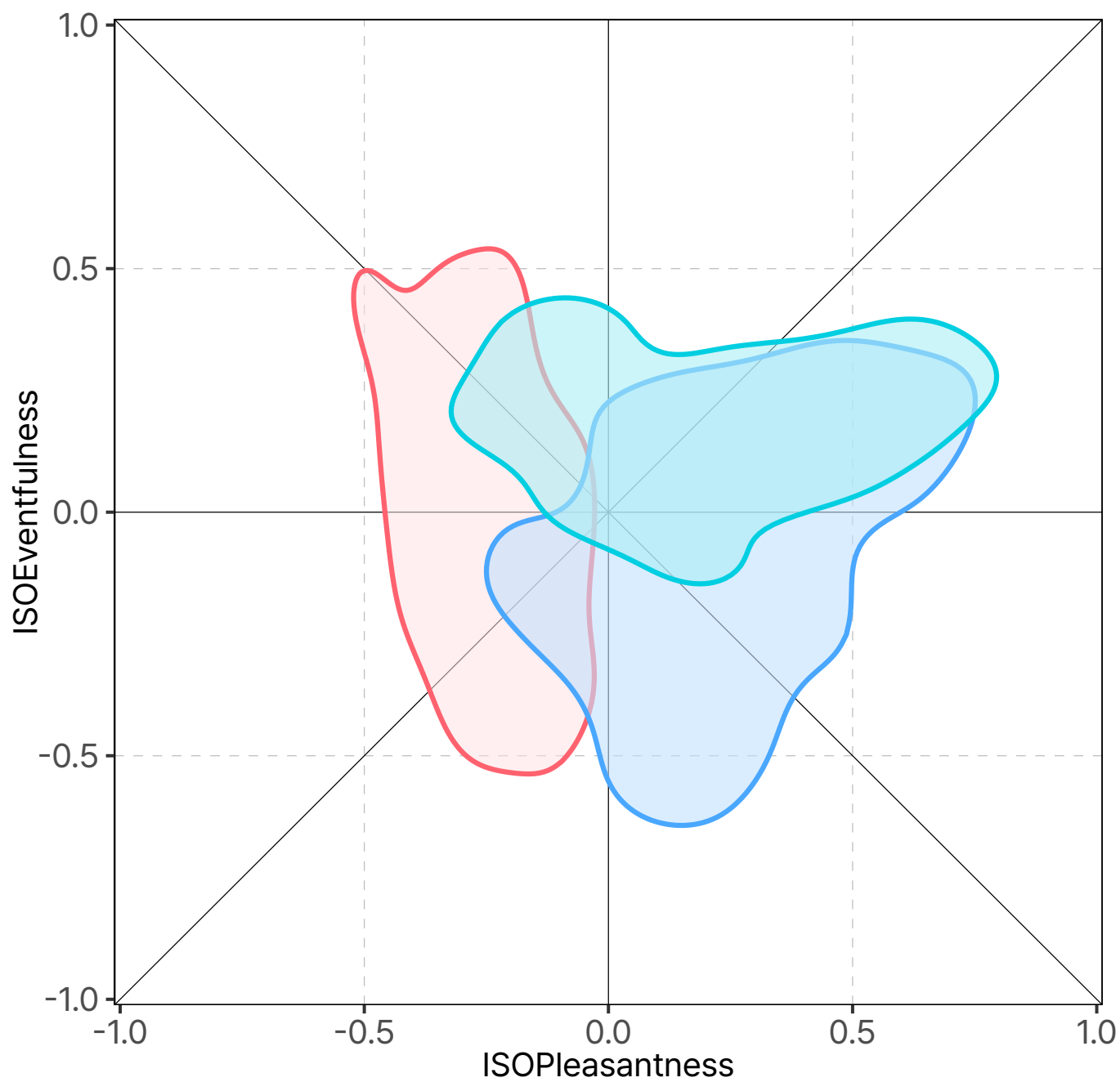












● Traffic noise (Hong & Jeon, 2020)
 ● Cluster 1 (D20)
 ● Cluster 1 (R6)